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A hybrid machine learning and IoT system for driver fatigue monitoring in connected electric vehicles

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ABSTRACT

Driver fatigue remains a critical factor in road accidents, particularly in long duration or cognitively demanding driving scenarios. This study presents a comprehensive, low cost, and real time system for monitoring driver health and electric vehicle status through physiological signal analysis. By integrating heart rate, eye movement, and breathing rate sensors, both simulated and real, this hybrid framework detects signs of fatigue using machine learning classifiers trained on publicly available datasets including OpenDriver, DriveDB, MAUS, YawDD, TinyML, and the Driver Respiration Dataset. The system architecture combines Arduino based hardware, cloud integration via Microsoft Azure, and advanced classification and anomaly detection algorithms such as Random Forest and Isolation Forest. Evaluation across diverse datasets revealed robust fatigue detection capabilities, with OpenDriver achieving 97.6% cross validation accuracy and 95.8% F1-score, while image and respiration-based models complemented the electrocardiogram-based analysis. These results demonstrate the feasibility of affordable, multimodal health monitoring in EVs, offering a scalable and deployable solution for enhancing road safety.

1. Introduction

With the rapid advancement of electric vehicles (EVs), autonomous driving technologies, and connected mobility platforms, ensuring driver safety has evolved from focusing solely on mechanical performance to encompassing real-time human physiological monitoring. (H. -T. Hsu, 2021) Despite breakthroughs in advanced driver assistance systems (ADAS), a major portion of road accidents still stems from human-related factors, particularly driver fatigue, cognitive overload, and reduced situational awareness. According to global road safety reports, fatigue related crashes remain alarmingly prevalent, especially in long haul driving, ride sharing operations, and semi-autonomous contexts where human supervision is intermittently required (Anna Skarbek-Żabkin, 2018).

As EVs continue to dominate the transportation landscape, the integration of intelligent, health-aware modules into vehicular systems has become a pressing priority (Y. Liu, 2019). Recent studies have emphasized the role of the Industrial Internet of Things (IIoT) in advancing electric vehicle ecosystems by enabling intelligent connectivity, predictive maintenance, energy optimization, and cloud-integrated analytics, thereby providing a foundational infrastructure for health-aware, AI-driven, and safety-centric EV applications (AlHousrya et al., 2025). In parallel, optimization techniques originally developed for renewable energy systems, such as PID-based metaheuristic algorithms for photovoltaic parameter extraction, have demonstrated high accuracy and computational efficiency, highlighting their potential applicability in energy-aware electric vehicle systems and smart charging optimization frameworks (Bennagi et al., 2025). Fatigue manifests in various forms: physical (muscle tiredness), cognitive (mental load), and ocular (eye strain), each influencing reaction times and decision-making ability (Rastgoo et al., 2024). Traditional ADAS solutions often overlook these subtleties, operating under the assumption that the driver remains perpetually alert and attentive (L. Yue, 2020).

Recent advancements in wearable devices and biomedical sensors have demonstrated the feasibility of continuously tracking physiological parameters such as heart rate variability, respiration, and ocular movement. However, the translation of these technologies into the automotive domain remains limited. Many existing prototypes constrained to laboratory environments or small-scale pilot studies, leaving

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a gap in scalable, real-world implementations. Furthermore, the unique driving context characterized by long durations, varying lighting conditions, motion artifacts, and diverse driver profiles demands specialized solutions that differ from traditional healthcare monitoring systems (Y. Liu, 2019).

In addition, the shift toward connected and shared mobility services introduces new opportunities and challenges for driver health monitoring. Fleet operators, ride-hailing companies, and logistics providers increasingly recognize the importance of monitoring driver well-being as a means of reducing accidents, lowering insurance costs, and improving service reliability. Current solutions either remain prohibitively expensive or fail to provide actionable insights in real time. Bridging the divide between low-cost sensor availability, machine learning–based inference, and cloud-enabled fleet management represents a crucial step toward achieving widespread adoption (Tzevelekakis, 2021).

Beyond the availability of sensors, the integration of artificial intelligence (AI) and Internet of Things (IoT) platforms have opened new horizons for real-time driver monitoring. Machine learning models, particularly ensemble methods and deep learning architectures, can extract subtle patterns from noisy physiological data, enabling robust fatigue classification across diverse scenarios (Trivedi, 2024). Coupled with IoT connectivity and cloud computing, these models allow for continuous data transmission, centralized analysis, and feedback loops that extend beyond the individual vehicle. Such integration transforms fatigue detection from a reactive, driver-only alert mechanism into a proactive, fleet-wide safety management tool, where anomalies can be flagged, interventions initiated, and long-term health analytics derived at scale (Shakir, 2024).

While notable progress has been made in driver monitoring research, several obstacles continue to hinder large-scale adoption. Many existing systems remain limited to single-modality sensing such as relying only on eye-tracking or ECG which reduces reliability in real-world driving where environmental noise and driver diversity are inevitable (A. Subasi, 2022). High hardware costs also restrict scalability, as most solutions depend on proprietary or specialized equipment not suited for mass deployment in personal vehicles or fleet operations. Furthermore, the integration of IoT and cloud capabilities, though increasingly recognized as essential for connected mobility, is still largely absent from current prototypes, preventing real-time fleet-level analytics and remote interventions (H. S. AlZu'bi, 2013). Finally, the issue of generalizability remains critical: methods validated in controlled laboratory settings often fail to transfer across diverse driver populations, varied fatigue types, and dynamic on-road conditions. These gaps underscore the need for a cost-efficient, multi-sensor, and cloud-connected solution that is robust, scalable, and adaptable to both experimental and real-world contexts (Khan, 2023).

1.1. Aims and objectives

This research addresses these critical gaps by developing and validating a modular, low-cost, multi-sensor driver health monitoring system capable of real-time fatigue detection. The system is specifically designed with scalability, affordability, and cloud integration in mind, making it well-suited for both personal EVs and large-scale fleet operations. The goals of this study are to:

- Capture multi-modal physiological signals (ECG, blink patterns, respiratory signals) using affordable, off-the-shelf components.
- Develop machine learning pipelines that combine rule-based and supervised approaches (Random Forest, CNNs) for robust fatigue detection.
- Ensure generalizability across physical, cognitive, and combined fatigue scenarios through real-world and experimental datasets.
- Enable IoT and cloud-enabled monitoring for real-time anomaly detection, fleet analytics, and remote interventions.

1.2. Research questions and contribution

Building on these goals, this study is guided by several key research questions. First, it asks how multi-sensor physiological monitoring can be effectively combined with machine learning techniques to detect different forms of driver fatigue in real time. A second question concerns the preprocessing and classification strategies required to ensure robustness against sensor noise, environmental variability, and individual driver differences. The research also investigates how IoT and cloud integration can support scalable, cost-effective, and remotely accessible monitoring systems suitable for both personal vehicles and fleet operations. Finally, it explores the extent to which the proposed system can generalize across both controlled experimental tasks, such as cognitive load induced by n-back tests, and real-world long-duration driving scenarios.

By explicitly addressing these gaps, this research contributes a comprehensive and deployable solution to the longstanding challenge of monitoring driver fatigue. It bridges physiological sensing, AI-based inference, and cloud-based deployment into unified, affordable, and scalable architecture. Ultimately, the proposed system lays a foundation for smart mobility ecosystems that are not only autonomous and electric but also health-aware, adaptive, and safety-centric.

2. Literature review

In recent years, driver monitoring systems (DMS) have transitioned from single-sensor designs, such as eye-tracking or steering behavior analysis, to more intelligent multimodal architectures that combine physiological, behavioral, and environmental inputs. Such systems are reported in literature to deliver more reliable detection of driver fatigue than approaches relying on a single data source. Electrocardiogram (ECG)–derived heart rate variability (HRV) features have gained traction as objective biomarkers for fatigue detection. In a study by Zeng et al., HRV features extracted from short (~2 min) ECG segments achieved high classification performance, while also considering sex-based physiological differences in model training and evaluation (Zeng C, 2024). Wavelet-based HRV analysis combined with machine learning reported for higher sensitivity and accuracy in distinguishing drowsy from alert states compared to traditional Fourier-based method (Li, 2013).

Another field-specific application reported LightGBM applied to HRV and respiratory rate features, enabling automatic three-level classification of flight fatigue with accuracy $\approx$ 88.6 %. This suite of studies underscores the efficacy of HRV features as reliable fatigue indicators, especially when combined with advanced machine learning models (Guo D, 2025). Visual fatigue detection relies on facial cues such as blink duration, yawning, and eyelid closure metrics like PERCLOS. Deep learning models trained in eye-region images and facial videos have attained high accuracy rates in detecting drowsy states. However, purely vision-based systems are vulnerable to low-light conditions, occlusions, and varying camera angles, which limits their reliability in real-world driving scenarios (Ojo, 2017). Many recent studies have demonstrated superior performance by fusing physiological and behavioral modalities. Integrated models combining HRV metrics (from ECG) with visual features such as PERCLOS significantly outperform unimodal counterparts, often achieving detection accuracy above 92 %. For example, multimodal systems that use joint embeddings from ECG, EEG, and facial video data have shown enhanced robustness across different data sources and driving conditions (Pillalamarri & Shanmugam, 2025).

Unsupervised methods such as Isolation Forest (IF) have become increasingly popular for health monitoring due to their ability to detect outliers in streaming data with linear runtime and minimal supervision (Heigl, 2021). Enhancements like stream-aware Isolation Forest variants

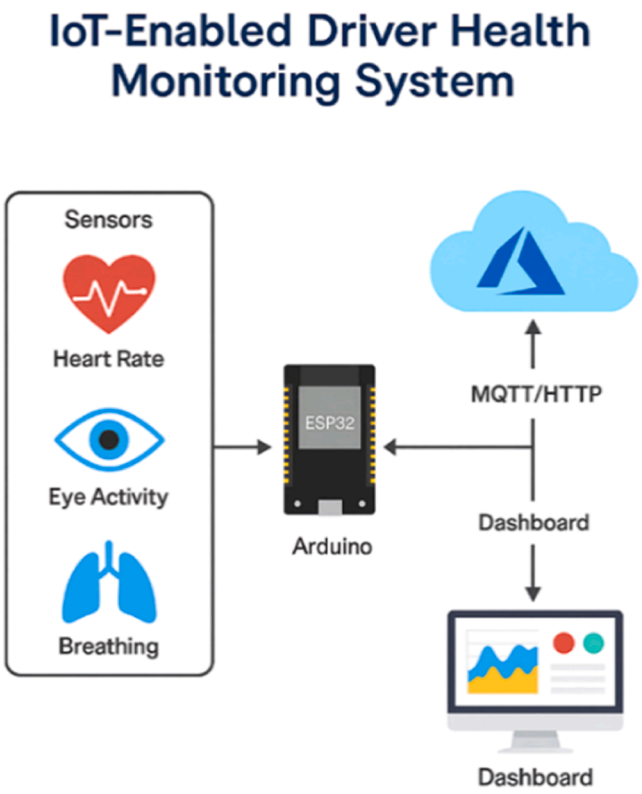


Fig. 1. System schematics of IoT enabled driver health monitoring.

have improved anomaly detection in real-time data pipelines. These methods are especially relevant in IoT-enabled health contexts, where continuous physiological monitoring generates high-volume temporal data (Ayad, 2025). Edge–cloud frameworks, using platforms such as Azure IoT Hub or AWS IoT Core, enable real-time data ingestion, processing, anomaly detection, and visualization in large-scale deployments (Borra et al., 2025). Lightweight Isolation Forest models deployed at the edge for real-time health anomaly detection have also been proposed while maintaining system efficiency and security (Xu, 2023). These approaches support scalable remote user interfaces, alerting mechanisms, and secure processing of physiological inputs.

Recent studies have increasingly focused on multimodal approaches to driver fatigue detection, recognizing that single-modality systems often suffer from limited robustness under real-world conditions (Abbas & Alsheddy, 2021). Combinations of physiological signals such as ECG, respiration, and electrodermal activity with visual cues including eye closure, blink dynamics, and facial expressions have been shown to improve fatigue recognition accuracy (Zhao et al., 2022). In parallel, several works have explored Internet-of-Things (IoT) and cloud-based architectures to enable real-time data acquisition, processing, and remote monitoring in intelligent transportation systems (Guerrero-Ibanez et al., 2015). These studies highlight the growing emphasis on scalable, low-cost, and connected fatigue monitoring frameworks, while also noting challenges related to sensor reliability, data heterogeneity, and generalization across subjects and driving environments (Mardanshahi et al., 2025).

Recently reported commercial driver monitoring solutions, such as Optalert and Ford BlueCruise, rely predominantly on visual heuristics or steering-based triggers and lack integrated physiological sensing and adaptive machine learning models. This gap highlights the need for scalable multimodal systems capable of combining heart rate, eye, and respiratory signals with anomaly detection, delivered via cloud-based architectures, as proposed in this work (Knapik, 2024).

3. Methodology

This section describes the technical methodology adopted for the development of the proposed driver health monitoring system. The focus is on sensor integration, physiological data acquisition, cloud-IoT architecture, and the machine learning-based analytics pipeline. The system is designed to monitor multiple health indicators in real-time, enabling both localized and remote detection of driver fatigue or anomalies across electric vehicle fleets.

3.1. Sensor integration and data acquisition

The foundation of the system is a network of simulated physiological sensors interfaced with microcontrollers to acquire real-time health data. Three primary health signals were targeted: heart rate, eye movement, and respiration.

The sensors were integrated using Arduino Uno microcontrollers configured for periodic signal sampling and preprocessing. In the setup, heart rate was simulated using a photoresistor that mimicked pulse signal variations, which were sampled through the analog input of the microcontroller (Tjitra, 2024). Eye movement was emulated using a potentiometer, representing eyelid motion by providing resistance-based analog readings corresponding to changes in eye position. For simulating breathing rate, an MQ-2 gas sensor was employed to detect fluctuations in gas concentration, effectively representing respiration cycles through periodic variations in its output signal (Kim, 2024). Each sensor was individually connected to Arduino's analog input pins (schematics of system are shown in Fig. 1). Signal sampling was performed at fixed intervals, and readings were converted into digital feature vectors using the onboard Analog-to-Digital Converter (ADC).

To support mobile and embedded deployment in vehicles, two primary communication protocols were employed. MQTT, a lightweight publish–subscribe protocol, was utilized for efficient real-time telemetry transmission, particularly suited for constrained network environments common in vehicular systems (Alshammari, 2023). Additionally, HTTP was used for structured data transmission via REST APIs, enabling reliable communication with cloud endpoints where broader compatibility and standardized integration were required. This combination ensured both low-latency data exchange and robust interoperability with cloud infrastructure. Both Wi-Fi modules (ESP8266) and cellular modules (SIM800L) were tested for network transmission depending on availability and mobility (P. W. Digarse & S. L. Patil, 2017). Data was transmitted via UART from the Arduino to connected edge devices before being relayed to the cloud.

To ensure data consistency across sensor modalities, all raw signals were normalized within defined operational ranges based on expected physiological behavior and sensor specifications. For heart rate simulation, the normalized range was set between 50 and 120 beats per minute (BPM) reflecting typical resting to elevated heart rate values encountered during driving scenarios (H. S. AlZu'bi, 2013). Eye movement signals, obtained from potentiometer-based resistance changes, were sampled through a 10-bit ADC (0–1023) and normalized to a 0–1 scale, corresponding to eyelid motion from fully closed to fully open positions. This range was selected to represent the full span of eyelid motion (M. S. Devi & P. R. Bajaj, 2008), from closed to fully open positions (Wen-Bing Horng, 2004). Similarly, the breathing rate signal derived from the MQ-2 gas sensor was captured as analog voltage levels (0–1023) and normalized to a 0–1 scale, representing variations in gas concentration corresponding to inhalation and exhalation cycles. This range captured the variations in gas concentration levels associated with inhalation and exhalation cycles (Schulman et al., 2024). Normalizing these signals ensured uniformity in downstream processing and enabled reliable comparisons across different sensing modalities.

The sensors employed in the prototype implementation were selected to emulate physiological signal patterns rather than to perform

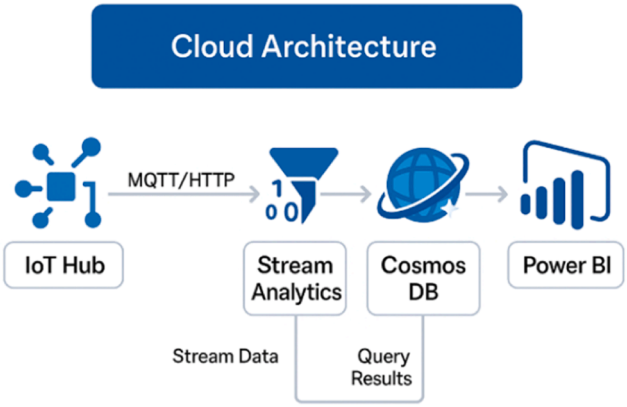


Fig. 2. Cloud architecture of Power BI dashboards.

direct biomedical measurement. Specifically, the photoresistor, potentiometer, and MQ-2 sensors function as signal proxies to validate data acquisition, edge-to-cloud transmission, and system interoperability. These components are not intended to replace clinical-grade sensors, and their use should be interpreted as a proof-of-concept demonstration of system architecture rather than a deployable medical solution. This system demonstrates a scalable architecture for real-time physiological data integration and fatigue-related pattern analysis.

3.2. Cloud-IoT architecture

To enable scalable, real-time processing and monitoring, the system was deployed on Microsoft Azure IoT infrastructure, which supported data ingestion, anomaly detection, alerting, and visualization (Y. Liu, 2019).

3.2.1. Real-Time data pipeline, storage and visualization

To enable seamless real-time monitoring, incoming feature vectors streamed through Azure IoT Hub using both MQTT and HTTP protocols. These protocols allowed efficient data transmission from vehicle-embedded microcontrollers to the cloud. Azure Stream Analytics was employed to continuously parse the incoming telemetry data and trigger processing jobs in real time (Park, 2017). For fatigue detection, an Isolation Forest (IF) model, trained offline on labeled physiological datasets, deployed via Azure Machine Learning Services. This model was responsible for classifying incoming signals as either normal or indicative of driver fatigue, enabling timely intervention (Zhao, 2023). All processed health telemetry and anomaly detection logs were stored in Azure Cosmos DB, a scalable NoSQL database service that supports high-throughput data ingestion and querying (Jeong, 2016). To provide stakeholders with actionable insights, Power BI dashboards developed as shown in Fig. 2.

The Power BI dashboard interface designed to present real-time analytics and historical trends in an intuitive layout. It included visual panels for heart rate, eye activity, and respiration signals, alongside live fatigue detection alerts generated by the Isolation Forest model. A summary panel displayed daily and weekly fatigue patterns, while another section provided fleet-level statistics for multi-vehicle monitoring. The data flow, illustrated in the system flowchart, shows how telemetry data streamed from Azure IoT Hub through Stream Analytics and Cosmos DB into Power BI for visualization, ensuring a continuous and responsive feedback loop. These dashboards offered real-time visualization of driver health states, historical fatigue trends, and detection summaries, making them valuable tools for both individual drivers and fleet managers overseeing multiple vehicles (Gonçalves, 2023).

Table 1
Evaluation metrics.

Test Scenario	Upload Latency (ms)	Detection Accuracy (%)	Alert Response (ms)	Observation
Calm Driving	180	95	250	Stable performance
Heavy Traffic	250	92	300	Slight delays under load
Night Driving	200	93	270	Consistent detection
Emergency Braking	150	94	220	Fast response to critical events

3.2.2. Security measures and alerting mechanism

To safeguard the integrity and confidentiality of the system, multiple security mechanisms implemented. All data transmissions between devices and cloud services secured using TLS/SSL encryption. Each device was authenticated using unique tokens and connection strings managed through Azure Device Provisioning Service (DPS), ensuring trusted communication (M. N. Bhuiyan, 2021). Furthermore, the IoT Hub enforced per-device access control policies, maintaining strict data isolation and preventing unauthorized access to telemetry from other devices in the network (Lakshmi Priya Rachakonda, 2024).The system is designed to support both local and remote alerting mechanisms to promptly notify relevant parties of any detected abnormal physiological conditions in drivers. Local alerts were activated directly within the vehicle using onboard components. When an anomaly is identified by the cloud-based detection model, signals are sent back to the microcontroller, which then triggers immediate feedback such as buzzer sounds or dashboard LEDs via GPIO outputs (Murat Arda Onsu, 2025). This ensures that the driver receives real-time warnings without dependence on external connectivity. In parallel, remote notifications were managed through cloud-based functions that disseminate alerts to connected mobile devices and monitoring dashboards. These notifications allow fleet managers and health supervisors to access real-time information about a driver's status, enabling them to take timely and informed actions when fatigue or other critical states are detected (Mohammed, 2014). This dual-layered alerting mechanism ensures both immediate driver feedback and centralized oversight.

3.2.3. Evaluation setup

To validate the responsiveness and robustness of the system, testing was conducted across varying driving scenarios. Key evaluation metrics are enlisted in Table 1which included data upload latency, anomaly detection accuracy, and alert responsiveness simulated through custom Python scripts and publicly available driver datasets that reproduced different driving conditions.

3.3. System architecture

The overall architecture of the proposed driver health monitoring system follows a modular, three-layered design that enables real-time physiological monitoring, cloud-based analytics, and timely feedback delivery. This architecture supports scalability, reliability, and low-latency operation, making it suitable for deployment across electric vehicle fleets (A. Zanella, 2014). Sensors and edge layers correspond to physiological signals, simulated via photoresistor (heart rate), potentiometer (eye movement), and MQ-2 gas sensor (breathing rate), were collected by Arduino-based IoT edge devices. These microcontrollers perform analog-to-digital conversion and transmit structured feature vectors using UART over MQTT or HTTP protocols via Wi-Fi or cellular module (M. N. Bhuiyan, 2021). The cloud analytics layer corresponds to Azure IoT Hub receives the incoming telemetry and then processed in real time using Azure Stream Analytics. Fatigue and anomaly detection is performed using a deployed Isolation Forest model via Azure Machine

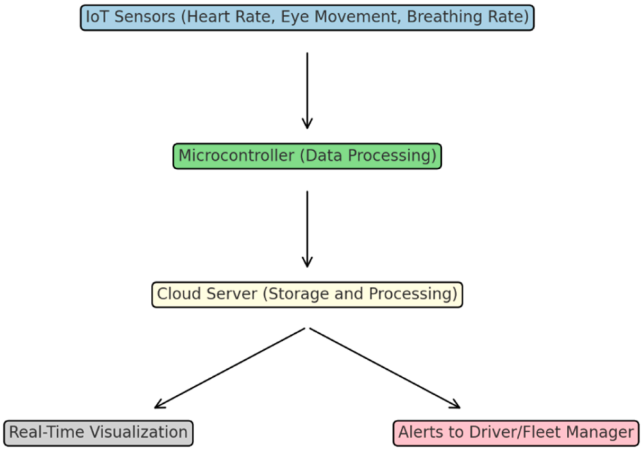


Fig. 3. Cloud architecture diagram.

Learning Services (T. Ahmed, 2021). All data is securely stored in Azure Cosmos DB, and Power BI dashboards, which offer live monitoring and historical insights (Medavarapu, 2023). Alerting and feedback detected anomalies trigger both onboard alerts (via LEDs or buzzers) and remote notifications sent to mobile devices or fleet dashboards. This dual-channel alert system ensures that both drivers and supervisors were informed in real time about critical physiological deviations. Fig. 3 in supporting information illustrates the full system architecture, highlighting data flow from sensor acquisition to cloud-based detection and feedback dissemination.

4. Datasets used

To evaluate the performance of the proposed multimodal fatigue detection system, we employed publicly available datasets spanning three physiological domains: ECG (cardiac activity), eye-based images (for blink and drowsiness detection) (Zhao et al., 2022) and respiratory signals (C. Yang, 2023). These datasets selected based on their relevance to driver monitoring scenarios, the diversity of recording conditions, and their compatibility with the modalities simulated and analyzed in this study.

4.1. ECG-Based fatigue detection datasets

Three distinct ECG datasets OpenDriver, DriveDB, and MAUS, were used to develop and validate the cardiac component of the fatigue detection pipeline.

4.1.1. OpenDriver dataset

The OpenDriver dataset comprises over 16 h of ECG recordings from eighteen drivers engaged in real-world driving tasks. This dataset was collected under naturalistic driving conditions, making it one of the most realistic and representative sources for physiological monitoring (Delong Liu, 2025). Each ECG signal is annotated with binary fatigue labels, derived through a combination of driver self-reports and expert evaluations (Delong et al., 2023). The signals exhibit a range of natural variability typical of in-vehicle conditions, such as changes in road type, traffic patterns, and driver posture. Notably, the dataset is well-balanced in terms of class distribution and features minimal noise, allowing for robust model training (Delong Liu, 2025). When evaluated using a Random Forest classifier, the dataset yielded high performance, with 97.6 % cross-validation accuracy and 96.4 % test accuracy.

4.1.2. DriveDB dataset

In contrast, the DriveDB dataset was recorded in a laboratory-controlled simulation environment, where 16 participants engaged in cognitive driving tasks while their ECG signals (Quentin Meteier, 2023) were monitored. The setup ensured consistent lighting, posture, and task complexity, yielding clean and artifact-free recordings. Fatigue labels were assigned based on task structure and subjective reports collected during transitions between cognitive load levels. Although the dataset enabled near-perfect classification performance (100 % test accuracy), it included a very small test set (only 6 samples), raising concerns about overfitting and generalizability to real-world conditions (M. Vaezi, 2024). Despite its limited size, DriveDB serves as a valuable benchmark for understanding fatigue under controlled cognitive stress.

4.1.3. MAUS dataset

The MAUS dataset introduces a more complex and varied setting, featuring over twenty individuals performing n-back tasks of increasing difficulty inducing mental fatigue. ECG recordings were collected in experimental sessions where participants were exposed to different cognitive workloads (Beh, 2021). Fatigue labels were assigned based on task intensity and subjective feedback. Compared to OpenDriver and DriveDB, the MAUS dataset exhibits greater inter-subject variability and was some class imbalance complicating model generalization. Nevertheless, the trained Random Forest model achieved 70 % cross-validation accuracy and 80.95 % test accuracy, highlighting the feasibility of fatigue detection even in cognitively induced stress conditions. The comparison of ECG-Based fatigue detection datasets is given in Table 2.

4.2. Eye-Based fatigue detection datasets

To capture visual indicators of fatigue such as blinking and eye closure, we utilized two image-based datasets: YawDD and the TinyML Eye dataset.

4.2.1. YawDD dataset

The YawDD contains video recordings of drivers under natural drowsy states. Frames were extracted from these videos and labeled as either “drowsy” or “alert” based on visual cues such as eye closure, yawning, and facial posture (Kır Savaş & Becerikli, 2023). The images reflect real-world variability in lighting, facial angles, and driver movements, making this dataset particularly suitable for training robust vision-based fatigue models. After preprocessing, including grayscale conversion and resizing, a convolutional neural network (CNN) was trained, achieving 93 % accuracy, precision, recall, and F1 score. The dataset’s diversity and realism contributed significantly to the model's ability to generalize across varying visual conditions (Abtahi, 2014).

Table 2
Comparison of ECG-based fatigue detection datasets.

Aspect	OpenDriver	DriveDB	MAUS
Recording Environment	Real-world driving	Laboratory simulation	Experimental cognitive tasks (n-back)
Subjects	18 drivers	16 participants	Over 20 participants
Signal Type	Raw ECG	Clean ECG under controlled setup	ECG during mental workload
Labeling Method	Self-reports + expert annotations	Task-based fatigue transitions	Task difficulty + subjective feedback
Strengths	Naturalistic setting, long-duration data, balanced labels	High signal clarity, structured tasks	Cognitive fatigue modeling, diverse participants
Limitations	Slight real-world signal variability	Very small test set, risk of overfitting	Inter-subject variability, some class imbalance
Model Performance	97.6 % CV accuracy, 96.4 % test accuracy	90 % CV accuracy, 100 % test accuracy	70 % CV accuracy, 80.95 % test accuracy

Table 3
Comparison of eye-based fatigue detection datasets.

Aspect	YawDD	TinyML Eye
Data Type	Video frames (grayscale) of real drivers	Grayscale images of eyes (cropped and minimalistic)
Application Setting	Natural driving with spontaneous drowsiness	Edge-device targeted, noise-free environment
Subjects	Multiple real-world drivers	Multiple individuals (non-driving context)
Labeling Method	Visual indicators (yawns, blinks, posture)	Binary labels: eye open or closed
Strengths	Realistic expressions, yawns and head tilt captured	Clean, high-SNR data, suitable for embedded systems
Limitations	Variable lighting and pose, less structured annotations	Lacks real-world complexity or full facial context
Model Performance	93 % accuracy, precision, recall, and F1 score	92 % accuracy, precision, recall, and F1 score

4.2.2. TinyML eye dataset

TinyML Eye dataset consists of minimalistic grayscale images focused solely on the eye region, with binary labels indicating open or closed states (Alajlan, 2023). The dataset was optimized for edge-device deployment and contains minimal background noise, offering a clean and controlled environment for training lightweight models. The same CNN architecture was applied, yielding 92 % accuracy across all evaluation metrics. While less varied than YawDD, TinyML Eye excels in scenarios requiring fast inference on low-power devices (R. Samanta, 2025). Comparison of Eye-Based Fatigue Detection Datasets are given in Table 3.

4.3. Respiration-Based dataset

For breathing signal analysis, we utilized the Driver Respiration Dataset, which includes respiratory recordings from drivers during various cognitive and physical tasks (F. Guede-Fernández, 2019). The dataset captures breathing patterns using a respiratory belt sensor, with signals reflecting different activity levels, from calm states to cognitively intense scenarios. Each sample is labeled based on task difficulty and corresponding physiological stress (H. -T. Hsu, 2021). The dataset posed challenges in terms of signal quality and inter-subject variability. Many samples exhibited overlapping characteristics between fatigue and non-fatigue states (Schulman et al., 2024) leading to higher false negative rates. Consequently, the respiration-based model underperformed relative to the ECG and eye-based systems, indicating that breathing signals alone may be insufficient for reliable fatigue detection. However, when used in combination with other modalities, they contribute complementary information that enhances overall system performance.

5. Experiments and implementations

This section presents the experimental setup and implementation of the complete multi-modal fatigue detection system, detailing how each module; ECG signal analysis, eye-based CNN classification, and respiratory anomaly detection was developed, integrated, and evaluated using real-world and simulated datasets.

5.1. Fatigue detection using ECG signals

The overall fatigue detection framework is shown in Fig. 4. It begins with raw ECG signal ingestion, followed by the extraction of statistical features, label assignment via a custom rule-based fatigue detection scheme developed in this study, and classification using a Random Forest algorithm (Salman, 2024). The modular design ensures that the same pipeline can be consistently applied to multiple datasets, promoting replicability and adaptability.

5.1.1. Signal preprocessing and feature engineering

To extract meaningful physiological patterns from ECG recordings, the raw signals, originally stored as string-based representations, first converted into numerical arrays to facilitate signal processing. This transformation enabled the application of structured preprocessing and feature engineering techniques aimed at capturing fatigue-relevant cardiac dynamics. The first step involved peak detection using the find peaks algorithm, which was applied to each ECG segment to identify R-peaks within the waveform (Manikandan & Soman, 2012). This allowed for the computation of RR intervals, representing the time between successive heartbeats, a critical metric in fatigue analysis. Following peak detection, a set of statistical features were computed from each ECG segment to summarize both temporal and amplitude-related signal characteristics. Key features included the Mean RR Interval (mean_rr),

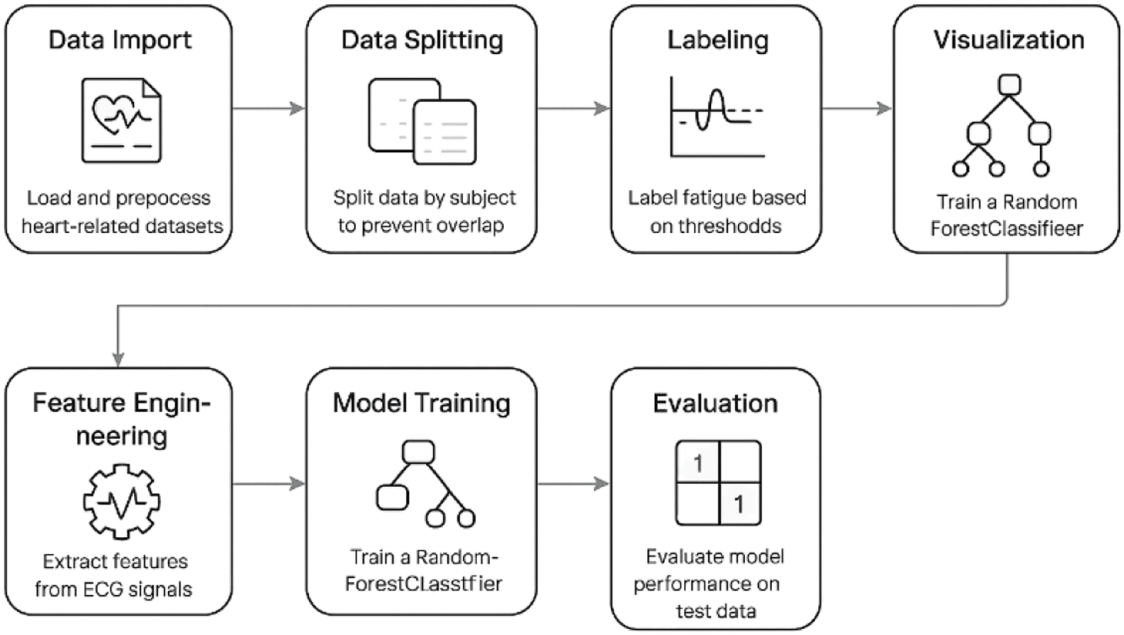


Fig. 4. Overview of the fatigue detection pipeline using ECG signals.

```
EGIN
 LOAD ECG data
 SPLIT data intoto train and test
OR each record DO
 EXTRACT features from ECG signal
 RETURN fatigue exceed thresholds
       RETURN fatigue
 ELSE RETURN not fatigue
PPLY fatigue labeling to train
     and test data
ET model = RandomForest
RAIN model on training data
REDICT on test data
ISPLAY confusion matrix,
     evaluation metrics
ND
```

Fig. 5. Pseudocode illustrating the logic for ECG signal processing, fatigue labeling, and classification using Random Forest.

which reflects the average duration between R-peaks, and the Coefficient of Variation of RR (rr_cv), a normalized measure of heartbeat variability. Additionally, the Standard Deviation of the Signal (signal_std) was calculated to capture general fluctuations in signal amplitude, which can indicate physiological stress or fatigue. Other features extracted included RR Range (rr_range), Number of Peaks (num_peaks), Signal Minimum (signal_min), Signal Maximum (signal_max), and Signal Mean (signal_mean). Together, these features provided a comprehensive summary of the ECG signal, encapsulated variations in both rhythm and amplitude that are known to correlate with fatigue-related physiological changes.

5.1.2. *Fatigue labeling strategy*

A rule-based labeling approach was employed to assign fatigue labels in the absence of explicit ground truth annotations, enabling the use of unsupervised physiological patterns for classification. Labels were derived based on the statistical distribution of selected training data features, ensuring consistency and objectivity in the absence of manual labeling. Specifically, each sample was labeled as Fatigued (1) if any of the three core features mean RR interval (mean_rr), coefficient of variation of RR intervals (rr_cv), or signal standard deviation (signal_std), fell below the first quartile (Q1), corresponding to the 25th percentile of their respective training distributions. This thresholding approach assumed that reduced variability and diminished signal dynamics are indicative of physical or cognitive fatigue. Samples that did not meet this criterion was labeled as non-fatigue (0). This method effectively flagged segments of ECG data that exhibited low variability or diminished cardiac responsiveness, both of which are commonly associated with fatigued physiological states. By relying on intra-distribution thresholds, the technique provided a scalable and reproducible strategy for labeling large volumes of unlabeled data in an unsupervised manner. The internal logic for ECG signal processing, feature extraction, labeling, and classification is summarized in the pseudocode shown in Fig. 5.

This structured code block outlines each step of the pipeline, beginning with the splitting of the dataset into training and test sets, followed by statistical feature extraction, label assignment based on the defined threshold conditions, and training of the Random Forest classifier. The pseudocode concludes with the evaluation phase, where model predictions were compared against the true labels using standard performance metrics. This clear representation ensures full transparency and supports the reproducibility of the methodology.

Table 4
Sensitivity of ECG labeling thresholds.

Threshold	Accuracy (%)	F1 Score
Q1 (25 %)	96.4	95.8
Q2 (50 %)	94.1	93.2
Q3 (75 %)	88.7	86.2

Reduced heart rate variability (HRV), prolonged RR intervals, and diminished signal amplitude have been consistently associated with fatigue-related autonomic imbalance, characterized by reduced parasympathetic activity and increased sympathetic dominance. Prior studies have reported that fatigue and sustained cognitive load led to lower RR variability and attenuated ECG dynamics, reflecting diminished cardiovascular adaptability (Lee et al., 2021). Motivated by these findings, the present study employs RR-interval–based statistical features as surrogate indicators of fatigue-related physiological changes. Due to the absence of explicit fatigue annotations in the utilized ECG datasets, a rule-based labeling approach was adopted to enable consistent and scalable annotation. Specifically, fatigue labels were assigned based on intra-distribution thresholds computed exclusively from the training data. Samples were labeled as fatigued when at least one of the core HRV-related features (mean RR interval, RR coefficient of

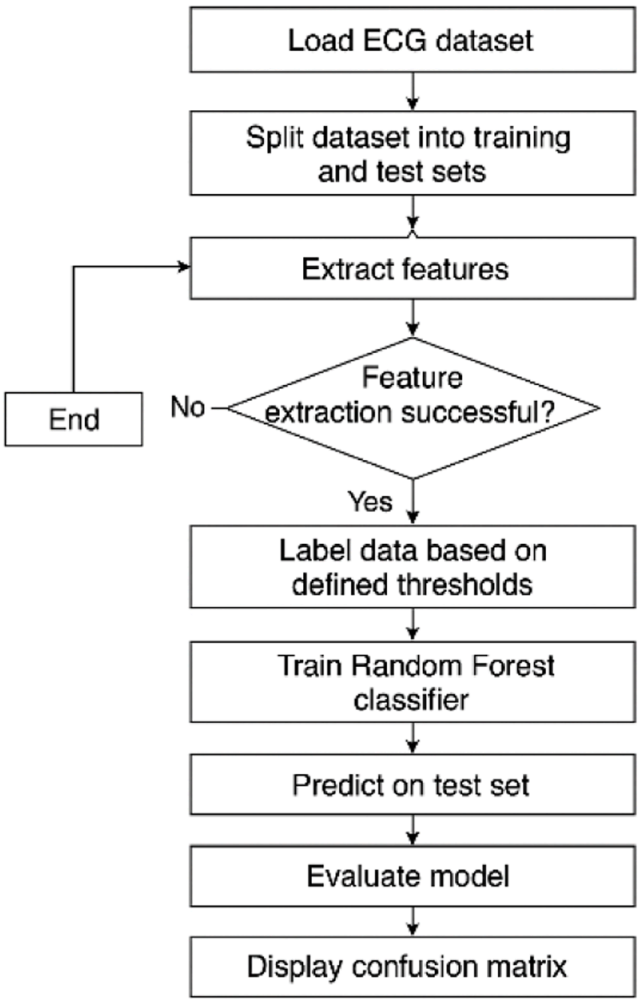


Fig. 6. Flowchart representing the complete pipeline from ECG.

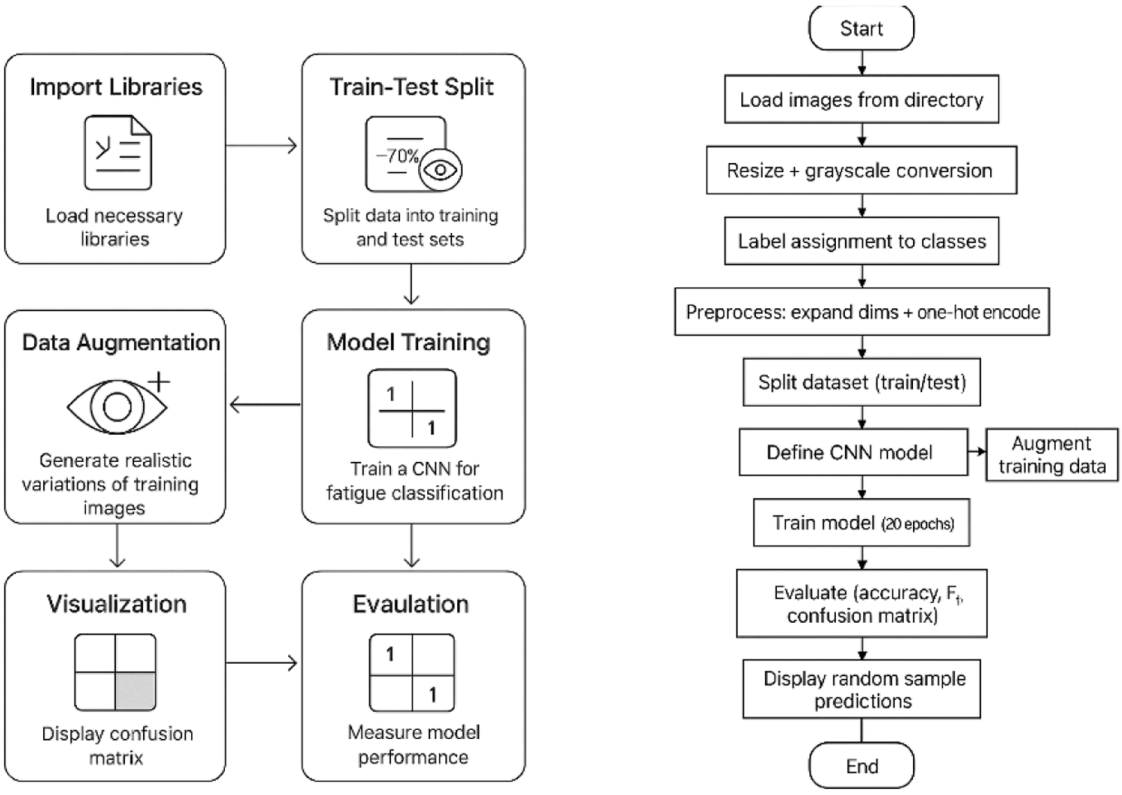


Fig. 7. CNN-based eye fatigue detection: from preprocessing to model evaluation (left), Flowchart of CNN training pipeline for fatigue detection using eye images (right).

variation, or signal standard deviation) fell below a predefined percentile threshold. This approach assumes that relative reductions in cardiac variability within a subject population correspond to fatigue-like physiological states.

5.1.2.1. Sensitivity analysis of labeling thresholds. To evaluate the robustness of the quartile-based labeling strategy, a sensitivity analysis was conducted by varying the percentile threshold used for fatigue assignment (Table 4). Thresholds corresponding to the 25th percentile (Q1), median (50th percentile), and 75th percentile (Q3) were tested. Model performance metrics remained stable for thresholds up to the median, while higher thresholds resulted in increased label noise and reduced classification performance. These findings indicate that lower-percentile thresholding provides a reasonable balance between sensitivity and specificity for unsupervised fatigue discrimination. Statistical aggregation and threshold sensitivity checks were assisted using AI-based analytical scripting tools.

It is important to note that the ECG fatigue labels employed in this study are heuristic and derived from relative physiological distributions rather than subjective or clinically validated ground truth. Consequently, the reported performance reflects the model's ability to discriminate relative fatigue-like patterns rather than absolute fatigue states.

5.1.3. Dataset splitting and training

To ensure fair evaluation and avoid subject-specific data leakage, each dataset was partitioned using the GroupShuffleSplit method. This splitting strategy ensures that all samples from a single subject appear in only one of the train or test sets, thereby promoting generalization and minimizing bias during evaluation. The classification model used in this pipeline is a Random Forest, which is known for its ability to manage noisy physiological signals, imbalanced datasets, and high-dimensional feature spaces. Each dataset was subjected to five-fold cross-validation on the training partition to assess the stability of model performance, followed by final evaluation on the independent test set using accuracy, F1-score, and confusion matrix analysis. To visually represent the end-to-end operation of the system, a flow chart is provided in Fig. 6. This diagram outlines all major functional steps beginning with raw ECG signal ingestion and proceeding through feature computation, label assignment, and model training. The flowchart also includes a conditional node that manages cases where feature extraction fails, ensuring that the pipeline remains robust during real-world deployment. The final node illustrates the evaluation phase, completing the pipeline with classification results and metric reporting.

5.1.4. Dataset-Specific implementation

To evaluate the robustness and generalizability of the proposed fatigue detection pipeline, we applied the same methodology across three publicly available ECG datasets: OpenDriver, DriveDB, and MAUS. Each data set represents a unique driving or cognitive workload condition, ranging from real-world urban driving to lab-controlled stress-inducing simulations. While preprocessing, feature engineering, and labeling strategies were uniformly applied, the datasets differed in signal context, experimental settings, and subject variability. The following subsections outline the structure, preprocessing, and labeling approaches specific to each dataset.

5.2. Fatigue detection with OpenDriver, DriveDB and MAUS dataset

For OpenDriver dataset, each ECG signal was first converted into a numerical array to enable computational analysis. RR intervals then extracted using the find_peaks function, which identified R-peaks within the ECG waveform and allowed precise measurement of the intervals between successive heartbeats. A comprehensive set of statistical features were computed for each signal segment. For fatigue labeling, three core features were extracted: the mean RR interval (mean_rr), which

reflects the average time between heartbeats; the coefficient of variation of RR intervals (rr_cv), indicating normalized heartbeat variability; and the standard deviation of the ECG signal (signal_std), capturing general amplitude fluctuations (Rao &, Martis). In addition to these, further features were computed to support model classification. These included the RR interval range (rr_range), the number of detected R-peaks (num_peaks), as well as the minimum, maximum, mean, and standard deviation of the ECG signal. These features provided a holistic view of both temporal and amplitude-based signal dynamics. To assign fatigue labels, a rule-based approach was applied. If any of the three core features, mean_rr, rr_cv, or signal_std, fell below the 25th percentile (first quartile) of their respective distributions in the training set, the corresponding segment was labeled as fatigue. Otherwise, the segment was labeled as non-fatigue. This unsupervised thresholding technique enabled consistent and scalable annotation of ECG data in the absence of manually labeled ground truth.

For DriveDB dataset, RR intervals were extracted from each ECG segment using peak detection. The same feature set applied in the OpenDriver dataset was used for consistency across datasets. For fatigue labeling, the features included mean_rr, rr_cv, and signal_std. For classification purposes, additional features such as rr_range, num_peaks, and the minimum, maximum, mean, and standard deviation of the ECG signal were computed. The labeling criteria remained identical to those used with the OpenDriver dataset. Any segment in which one or more of the labeling features, mean_rr, rr_cv, or signal_std, fell below the 25th percentile of its respective training-set distribution was marked as fatigue.

The feature engineering process for the MAUS dataset followed the same core methodology. RR interval-based features included mean_rr, rr_cv, and rr_range, capturing heartbeat timing and variability. Signal-based features were also extracted, including signal_std, signal_min, signal_max, and num_peaks, providing a quantitative view of amplitude and morphological characteristics of the ECG waveform. Fatigue labeling remained consistent with the approach used in the previous datasets, relying on threshold-based rules derived from training-set distributions of mean_rr, rr_cv, and signal_std. Additionally, the trial_type column in the dataset, which indicated the level of cognitive load (0-back, 2-back, or 3-back), was preserved to facilitate downstream subgroup analysis and enable visualization of how varying cognitive demands influence physiological fatigue responses.

5.3. Eye-Based fatigue detection using CNN

Visual cues such as prolonged eyelid closure, reduced blink rate, and yawning are reliable indicators of fatigue. Eye-based fatigue detection offers a non-contact, camera-based solution that is particularly suitable for real-time applications in vehicles. This section presents a CNN-based pipeline trained on two benchmark datasets, YawDD and TinyML Eye to classify eye images as fatigued or non-fatigued (Alimoon et al., 2025). The approach involves standardized preprocessing, robust CNN architecture design, and evaluation using relevant visual fatigue metrics (Makhmudov, 2024). An overview is given in Fig. 7 (left) and a flow chart for CNN training pipeline is in Fig. 7 (right).

5.3.1. Dataset description, preprocessing and augmentation

Two publicly available datasets were used for eye-based fatigue classification. For YawDD dataset (Shabnam Abtahi, 2020), the labels were annotated based on observable indicators such as yawning, partially closed eyes, or drooping eyelids. In contrast, for the TinyML Eye dataset, each image is labeled as either “tired” or “not tired,” making this dataset particularly suitable for training lightweight models intended for edge deployment (Alajlan, 2023). All images were converted to grayscales to reduce dimensionality and simplify feature extraction. They were then resized to 48 × 48 pixels to balance detail retention with computational efficiency. Pixel values were normalized to a 0–1 scale, and the images reshaped to match the input dimensions required by the CNN architecture. In addition to preprocessing, data augmentation techniques were employed during training to enhance model generalization and reduce overfitting. These augmentations included random horizontal flipping to simulate changes in head orientation, brightness variations to reflect diverse lighting conditions, and minor zoom transformations to account for slight differences in camera distance or eye framing. Together, these techniques mimicked the visual variability encountered in real driving environments. The complete preprocessing workflow is illustrated in Fig. 7 (right).

5.3.2. Dataset splitting and CNN model architecture

A stratified split strategy was employed to preserve the original label distribution across the different subsets of data. Specifically, each dataset was divided into 70 % for training, 15 % for validation, and 15 % for testing. This approach ensured that the ratio of fatigued to non-fatigue samples remained consistent across all phases of model development. Separate train–validation–test splits were created for both the YawDD and TinyML Eye datasets. This separation allowed each model to be evaluated independently while maintaining consistent label proportions, thereby supporting fair performance comparison and robust generalization analysis. The CNN architecture was optimized for binary classification of grayscale eye-region images and designed to balance computational efficiency with classification accuracy. The network began with two convolutional layers, each followed by ReLU activation to introduce non-linearity and enable the model to learn spatial features relevant to eye states (Alzubaidi et al., 2021). Max pooling layers were applied after each convolution to reduce the spatial dimensions of the feature maps, thereby decreasing computational load while retaining key information. A dropout layer with a rate of 0.2 was included to prevent overfitting by randomly deactivating neurons during training. The output of the convolutional stack was then passed through a flatten layer, which converted the 2D feature maps into a 1D vector suitable for dense processing. This vector fed into a fully connected dense layer with 128 units, allowing the model to learn high-level representations of the input. Finally, a sigmoid activation function applied in the output layer to produce a probability score for binary classification, distinguishing between fatigued and non-fatigue states. The overall architecture was lightweight and effective, making it suitable for deployment in both cloud environments and edge devices with limited computational resources.

5.3.3. Model training and evaluation strategy

Training conducted using a standard supervised learning pipeline tailored for binary classification. The binary cross-entropy loss function was used to measure the error between predicted and actual labels, and the Adam optimizer (Kingma, 2014) was selected for its efficiency and adaptability during gradient descent. The model was trained over 20 epochs with a batch size of thirty-two, allowing sufficient exposure to the data while maintaining manageable memory usage. To prevent overfitting, early stopping implemented based on validation loss, halting training if no improvement observed across consecutive epochs. Additionally, model checkpointing used to retain the best-performing model according to validation loss. Throughout training, both training and validation accuracy was monitored to assess convergence behavior and ensure generalization.

Model performance evaluated on the reserved test set using a comprehensive set of metrics. Accuracy measured the overall correctness of predictions, while precision quantified the proportion of correctly predicted fatigued cases among all positive predictions. Recall assessed the model's ability to detect actual fatigue cases in the data (Varoquaux & Colliot, 2022). The F1-score, as the harmonic mean of precision and recall, provided a balanced measure of the model's effectiveness, particularly in scenarios with potential class imbalance. These metrics offered a thorough evaluation of both the reliability and sensitivity of the fatigue detection system. Confusion metrices generated to further analyze true/false classifications.

```
images_train, images_val, images_test
labels_train, labels_val, labels_test
define define_cnn_model
COMPILE model with ()
WHILE epochs less > 20
CALL train_on_batch(imag_images_train, lab-
els_train)
CALL evaluate_on_batch(images_val,
labels_val)
SET batch_labels_test = CALLpredict_on_batc
(images_test)
DISPLAY accuracy, accuracy, preocision,
recall, ε1_score(data_test )
CALL calculate_accuracy(labells_test, batc-
labels_test)
CALL calculate_precision(labells_test,
batch_labels_test)
CALL calculate_recall(1labels_test,
batch_labels_test)
CALL generate_εf1_score(labels_test, batch_
labels_test)
DISPLAY call confusion_matrix
DISPLAY confusion_matrix DISPLRY accuracy
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DISPLAY confusion_matr
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```

Fig. 8. Pseudocode with key steps in CNN-based training and evaluation for fatigue detection.

Table 5
Overview of random forest hyperparameters and CNN architecture summary.

Random Forest Hyperparameters		CNN Architecture Summary	
Parameter	Value	Layer	Output
Trees	100	Conv2D	32
Max Depth	None	MaxPool	-
Class Weight	Balanced	Conv2D	64
Split Criterion	Gini	Dense	128
		Output	Softmax

An evaluation conducted independently for the YawDD and TinyML Eye datasets to understand dataset-specific model behavior. Sample predictions, including misclassified examples, are shown in Fig. 8 while the overall logic flow of the system is summarized in Fig. 8. The pseudocode outlines a deep learning pipeline for training and evaluating a CNN model in a batched workflow. The process involves dataset initialization (*train/val/test* splits), model definition, and iterative training over 20 epochs with batch-wise updates (*train_on_batch*) and validation (*evaluate_on_batch*). Test-set predictions were assessed using standard metrics (accuracy, precision, recall, F1) and visualized via a confusion matrix. While the structure adheres to best practices, separating data phases and emphasizing batch processing, minor inconsistencies (e.g., repeated commands, typos) suggest a draft version. The pipeline provides a scalable framework for model development and performance analysis. An overview of random forest hyperparameters and cnn architecture summary is provided in Table 5.

5.4. Fatigue detection using breathing-based features

Dataset Description and Preprocessing: To classify fatigue using respiratory-based features, a structured machine learning pipeline implemented. The dataset comprised pre-engineered physiological variables derived from respiration and heart rate variability (HRV), with fatigue labels (label_sleep) already provided, eliminating the need for manual annotation. Since multiple samples collected from the same individuals, data splitting was handled using the GroupShuffleSplit strategy based on subject_id, ensuring subject-independent training and testing sets. A Random Forest classifier selected due to its effectiveness in handling complex, nonlinear physiological data. The model trained on the full feature set, and performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score, to comprehensively assess its ability to distinguish fatigue from non-fatigue states.

5.5. Analytical workflow and model inputs

Fig. 9 presents a consolidated overview of the complete analytical workflow employed in this study, spanning signal acquisition, segmentation, feature extraction, labeling, and model inference. Although each modality differs in signal characteristics, a unified design philosophy adopted to ensure consistent preprocessing and reproducibility. For all modalities, raw signals were segmented into fixed-duration windows, from which task-relevant features or representations extracted and subsequently used as inputs to machine learning models (Table 6). For ECG and respiration modalities, each observation corresponds to one segmented signal window represented by a fixed-length statistical feature vector. For the vision modality, each observation corresponds to an eye-region image frame or short image sequence processed directly by a convolutional neural network. This explicit definition ensures that each model input corresponds to a clearly defined physiological observation.

5.5.1. Signal segmentation and observation definition

Continuous physiological recordings segmented into fixed-length windows to enable standardized feature extraction and model training (Table 6). Each segmented window constitutes a single observation used for classification. Window lengths were selected based on prior literature and signal-specific temporal dynamics, balancing physiological relevance with statistical robustness. Overlapping windows were employed for ECG and respiration signals to preserve temporal continuity and increase the number of training observations while minimizing information loss at window boundaries. Thus, each observation

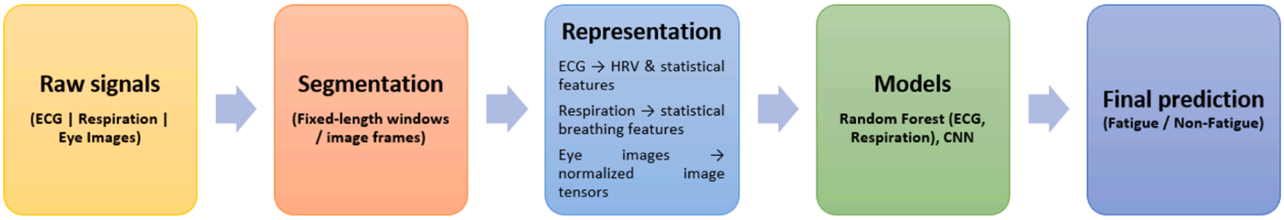


Fig. 9. A consolidated data analysis workflow.

Table 6
Model inputs, feature definitions and segmentation strategy across modalities.

Modality	Input Type	Extracted Variables / Inputs	Classification Model	Window Size	Overlap	Observation Definitions
ECG	Time-series windows	mean_rr, rr_cv, rr_range, num_peaks, signal_min, signal_max, signal_mean, signal_std	Random Forest	30s	50 %	One HRV feature vector per window
Respiration	Time-series windows	mean amplitude, variance, breathing rate, signal entropy	Random Forest	30s	50 %	One statistical feature vector per window
Eye Images	Image frames	Grayscale eye-region images (normalized, resized)	CNN	Frame Level	N/A	One image or frame sequence

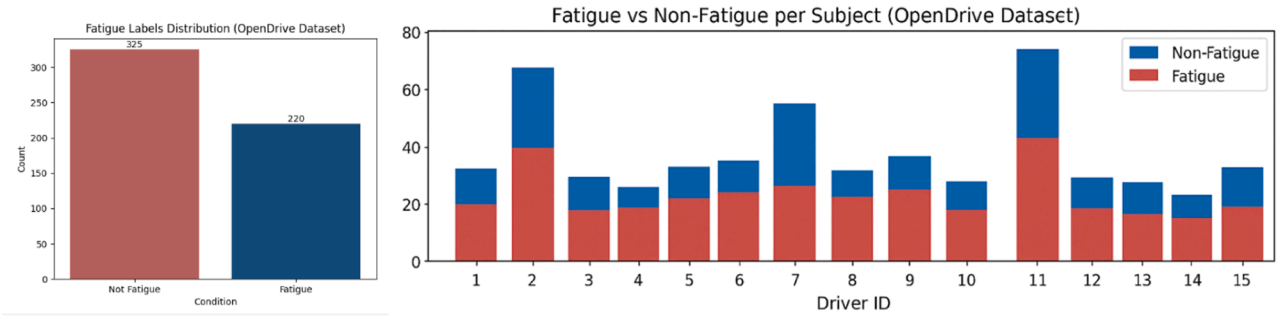


Fig. 10. Distribution of fatigue vs. non-fatigue labels (left), sample count of fatigue and non-fatigue conditions per driver (right) in OpenDriver dataset.

used in the Random Forest models corresponds to a single segmented physiological window summarized by a fixed-dimensional feature vector, while each CNN observation corresponds to a preprocessed eye-region image.

To improve methodological clarity, a consolidated data analysis workflow is presented in Fig. 9. Unlike system-level architecture diagrams, this figure focuses exclusively on the transformation of raw physiological signals into model-ready inputs and subsequent fatigue prediction. All continuous signals are segmented into fixed-length observations to ensure consistency across modalities. As illustrated in Fig. 9, each modality follows a dedicated processing pipeline. ECG signals undergo RR-peak detection and heart-rate-variability–based feature extraction, while respiration signals are summarized using statistical breathing descriptors. Eye-based fatigue detection operates directly on preprocessed image tensors using a convolutional neural network. The complete set of variables and model configurations employed for each modality is summarized in Table 6.

6. Results and discussion

6.1. Dataset 1: fatigue detection with the opendriver dataset

The Random Forest Classifier was trained on statistical features derived from preprocessed ECG signals. Model validation was performed using 5-fold cross-validation (Kohavi, 1995), a method that partitions the dataset into five subsets to iteratively train and test the model, thereby minimizing the risk of overfitting and providing a robust estimate of performance as shown in Fig. 10 (right). As depicted in Fig. 10 (left), the training set was well balanced, consisting of 322 non-fatigue and 228 fatigue samples. This near-uniform distribution helped ensure that the Random Forest model learned representative patterns from both classes, reducing bias and improving generalization across conditions. In addition, Fig. 5 shows the subject-wise breakdown of sample labels, indicating that most drivers contributed both fatigue and non-fatigue instances.

This subject-level variability is critical for training models that can accommodate diverse physiological baselines and driving behaviors. The classifier demonstrated robust performance both during training and on the independent test set, indicating effective generalization and stability. The mean cross-validation (CV) accuracy across training folds

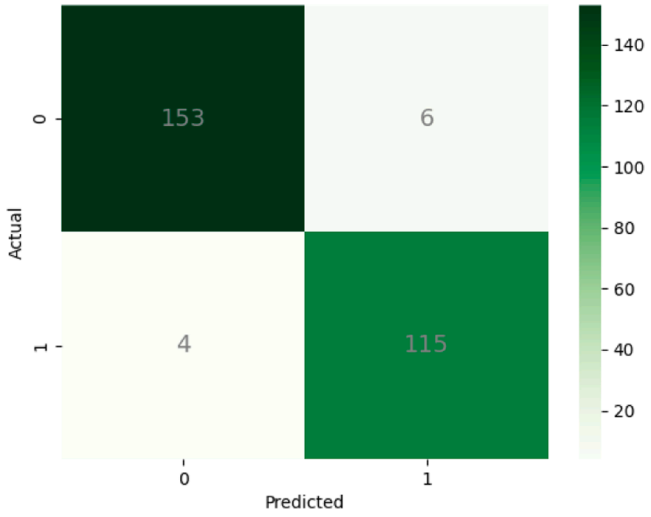


Fig. 11. Confusion matrix showing prediction performance on the OpenDriver test set.

was 97.6 %, with a low standard deviation of 0.73 %, reflecting consistent performance across data splits. On the held-out test set, the model achieved a test accuracy of 96.4 %, alongside an F1 score of 95.8 %, confirming balanced performance across both fatigue and non-fatigue classes. These metrics highlight the model's ability to learn meaningful representations from images from eye movements while maintaining stability across validation folds. The classification report further supports this by showing that both classes predicted with high precision and recall, demonstrating the model's reliability in distinguishing between drowsy and alert driver states. A breakdown of class-wise performance on the test set shows that both precision and recall exceed 0.95 for each class, confirming the model's reliability in detecting fatigue as well as accurately identifying non-fatigued states. For the non-fatigue class, the model achieved a precision of 0.97 and a recall of 0.96, indicating strong specificity and a low false positive rate. For the fatigue class, precision was 0.95, and recall reached 0.97, reflecting the model's effectiveness in capturing true fatigue instances

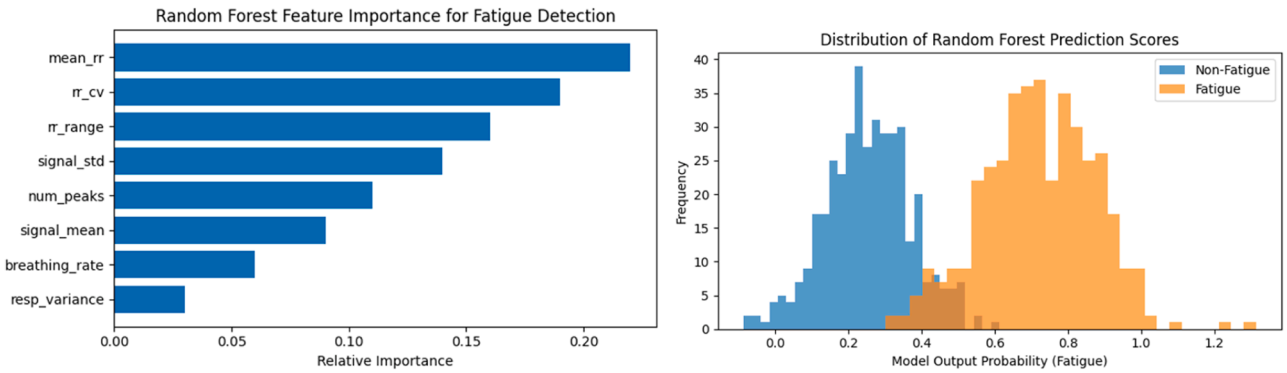


Fig. 12. Random forest feature importance for fatigue detection (left), Distribution of random forest prediction scores (right).

with minimal false negatives. These results demonstrate that the classifier is well-balanced, exhibiting no significant bias toward either class. Its high sensitivity and specificity make it a robust tool for real-world deployment in fatigue detection systems. The confusion matrix analysis is shown in Fig. 11.

The model correctly identified 115 fatigue cases as true positives and accurately classified 153 non-fatigue cases as true negatives. There were 6 false positives, where non-fatigue samples incorrectly labeled as fatigued, and 4 false negatives, where actual fatigue instances missed. The small number of misclassifications (total of 10) supports the reliability of the classifier. The false negative rate is especially important in the context of driver fatigue detection, where undetected fatigue could lead to safety-critical incidents. The model's high recall for the fatigue class (0.97) ensures that most fatigue cases were correctly detected. Overall, the results from the OpenDriver dataset confirm that ECG-derived statistical features, when processed through the proposed pipeline, can accurately distinguish between fatigue and non-fatigue states in naturalistic driving conditions.

Beyond aggregate accuracy metrics, model behavior can be interpreted by examining class-wise error patterns reflected in the confusion matrices. For the ECG-based model, balanced true positive and true negative rates indicate that heart rate variability features provide stable discrimination between fatigue and non-fatigue states. This suggests that the classifier captures physiologically meaningful patterns associated with autonomic regulation under fatigue. In contrast, the respiration-based model exhibits a pronounced false-negative bias, indicating that breathing-related features often fail to reflect fatigue-induced changes consistently across subjects. The eye-based model demonstrates comparatively balanced error distribution, reflecting the strong association between ocular dynamics and fatigue-related visual cues. These modality-specific error characteristics provide insight into the relative reliability and limitations of each sensing modality without requiring additional model complexity or post hoc explainability techniques. As presented in Fig. 12 (left), relative feature importance derived from the Random Forest classifier. Heart-rate–variability–based ECG features dominate the decision process, while respiration-related features contribute marginally. This confirms that cardiac dynamics provide the strongest fatigue-related information, whereas breathing signals play a secondary role. Distribution of Random Forest output probabilities for fatigue and non-fatigue classes are presented in Fig. 12 (right). While the two distributions are largely separable, a region of overlap remains, explaining observed misclassifications. This visualization provides insight into classifier confidence and decision ambiguity beyond confusion matrices.

6.2. *Dataset 2: fatigue detection using the drivedb dataset*

The Random Forest classifier was trained using statistical features extracted from ECG signals, with model validation performed via 5-fold cross-validation. This approach ensures robust performance evaluation while minimizing the risk of overfitting. However, due to the limited size of the DriveDB dataset, the reliability of performance estimates remains constrained. The training data, illustrated in Fig. 13 (left), comprises 7 fatigue and 4 non-fatigue samples. While the class distribution is balanced, the absolute sample count is critically low, raising concerns regarding the statistical power of the model and the generalizability of the findings. Fig. 13 (right) presents the subject-level distribution of fatigue and non-fatigue labels. All participants contributed samples from both classes, which is methodologically beneficial for reducing subject-level bias during training. Nevertheless, the extremely small number of samples per subject limits the model's ability to capture inter-individual variability, which is essential in real-world fatigue detection scenarios.

The classifier exhibited strong quantitative performance on the second dataset, achieving excellent results across both cross-validation and independent testing. The mean cross-validation accuracy was 90.0 %, though accompanied by a high standard deviation of 20.0 %, indicating

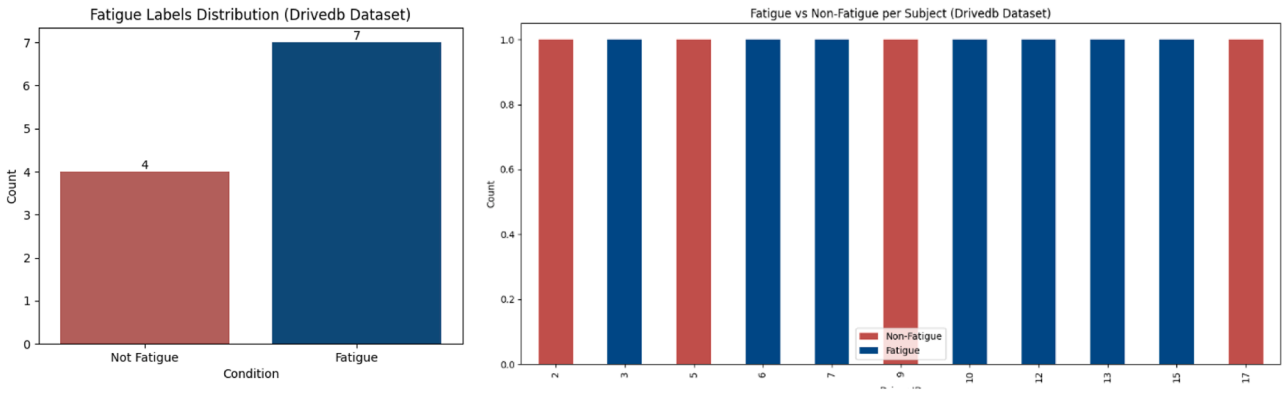


Fig. 13. Label distribution (left) and sample distribution per subject (right) of fatigue and non-fatigue in the DriveDB training data.

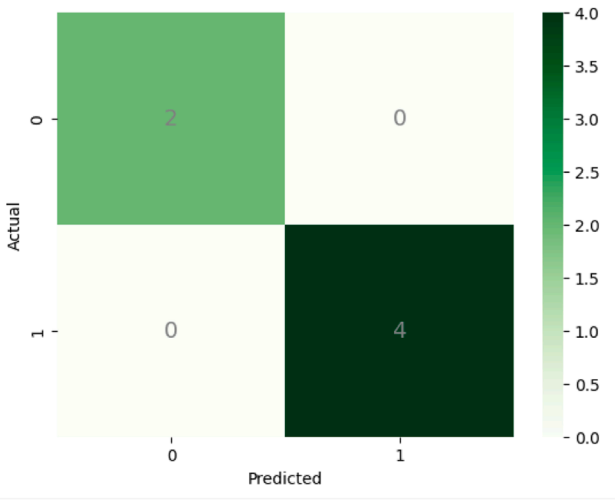


Fig. 14. Confusion matrix for fatigue classification on the DriveDB test set.

Table 7
Subject-wise Performance on DriveDB.

Subject	Accuracy (%)	F1 Score
S1	1.00	1.00
S2	0.83	0.80
S3	1.00	1.00
Mean ± SD	0.94 ± 0.09	0.93 ± 0.11

variability across different folds, due to the smaller dataset size or imbalanced distributions. Despite this variability during training, the model achieved perfect test accuracy of 100 % and an F1 score of 100 %, demonstrating its ability to generalize effectively on unseen data. Despite the favorable metrics, the notably high standard deviation in cross-validation highlights instability due to the limited data volume. Furthermore, the perfect test set performance must be interpreted cautiously, as it is based on only six samples. The classification report indicates flawless separation of the two classes on the test set. For the non-fatigue class, both precision and recall were 1.00, reflecting perfect identification of all non-fatigue instances without any false positives. Similarly, the fatigue class also achieved a precision and recall of 1.00, indicating the model correctly identified all fatigued samples without any false negatives. While these results are highly encouraging and suggest that the model effectively captured class-specific patterns, their statistical significance is constrained by the small size of the test set. The absence of misclassifications, although promising, does not guarantee generalizability to broader, more diverse populations. Further evaluation of larger and more heterogeneous datasets is essential to confirm the model's robustness in real-world conditions. The confusion matrix, shown in Fig. 14, confirms correct classification of all instances in the test set, comprising four fatigue and two non-fatigue samples. No false positives or false negatives observed.

The model correctly identified 4 fatigue instances as true positives and 2 non-fatigue instances as true negatives. There were 0 false positives and 0 false negatives. The complete absence of misclassifications reinforces the model's discriminative ability on this dataset. However, the extremely limited test size precludes any conclusive judgment regarding real-world performance. The results obtained from the DriveDB dataset demonstrate the potential of ECG-derived statistical features in classifying fatigue states under controlled conditions. The model achieved perfect performance on the available test set and high mean accuracy during cross-validation. Importantly, the classifier exhibited a recall of 1.00 for the fatigue class, indicating no fatigue cases missed, an essential requirement for safety-critical applications such as driver monitoring.

Although the DriveDB dataset contains a limited number of samples, it was not used in isolation to support the conclusions of this study. Instead, DriveDB was evaluated alongside substantially larger and more diverse datasets, including OpenDriver and YawDD, which contain extensive real-world driving recordings under varied operational conditions. Performance trends observed on DriveDB (Table 7) were consistent with those obtained on these larger datasets, particularly with respect to the discriminative capability of ECG- and vision-based modalities. Consequently, DriveDB results are interpreted as a supplementary validation case that demonstrates methodological consistency rather than as a primary source of generalization claims. The main conclusions of this work are therefore driven by evaluations on larger-scale and heterogeneous datasets, with DriveDB serving to highlight feasibility under constrained data settings.

Future studies should prioritize larger-scale datasets with higher subject diversity to enable statistically reliable subject-independent validation strategies such as leave-one-subject-out cross-validation.

6.3. *Dataset 3: fatigue detection using the MAUS dataset*

The Random Forest classifier was trained and validated on the MAUS dataset using 5-fold cross-validation. Compared to previous datasets, MAUS introduces greater intra-class variability due to its cognitive load-based experimental setup and inherent subjectivity in fatigue responses. Although the dataset is well-structured, its variability presents a more challenging environment for fatigue detection based on ECG-derived features. The overall label distribution in the training set was well-balanced, comprising 46 non-fatigue and 44 fatigue samples, as shown in Fig. 15 (left). This party supports unbiased model training across classes. However, the uniform class distribution at the dataset level does not compensate for imbalances observed at the subject level. As

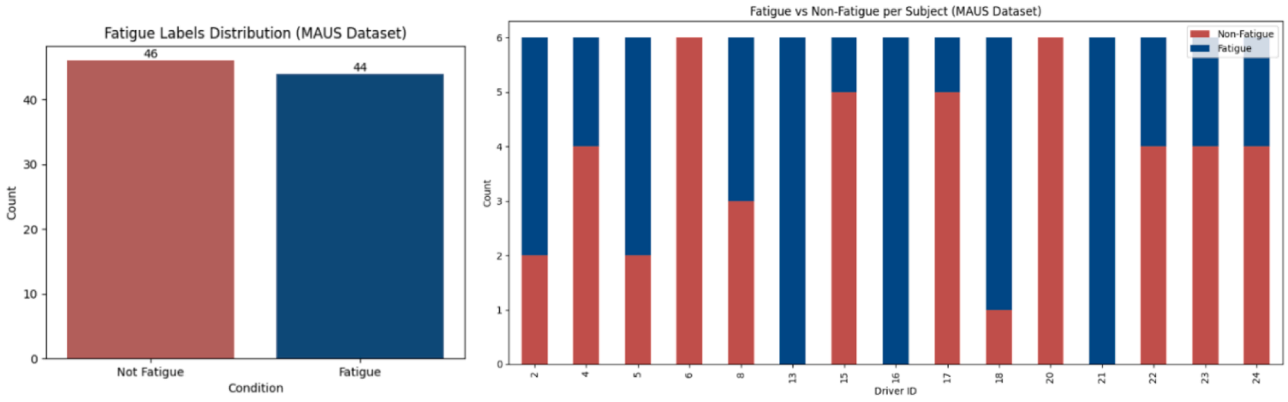


Fig. 15. Distribution of fatigue and non-fatigue labels (left) and sample distribution across subjects (right) in the MAUS training set.

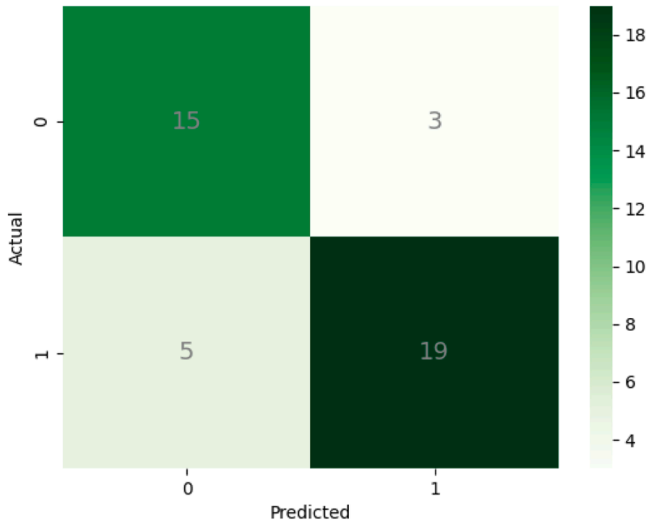


Fig. 16. Confusion matrix of test performance of fatigue classification in MAUS dataset.

illustrated in Fig. 15 (right), the subject-wise sample distribution shows noticeable skew across individuals, with some participants contributing predominantly to one class. This imbalance affects the model's ability to generalize and may partially explain reduced classification performance, particularly in recall for the fatigue class. Subjects with minimal or no representation in one class offer limited discriminatory value during training.

Despite these challenges, the model achieved the following performance. The mean cross-validation accuracy was 70.0 %, with a standard deviation of 9.7 %, reflecting moderate variability across folds. On the test set, the model achieved an accuracy of 80.95 % and an F1 score of 82.61 %. These results indicate moderate but promising classification performance, especially considering the high degree of subject variability and cognitive task-induced noise present in the ECG recordings. The class-wise metrics for test performance are as follows. For the non-fatigue class, the model achieved a precision of 0.75 and a recall of 0.83, indicating reasonably superior performance in correctly identifying non-fatigued instances. For the fatigue class, the precision was 0.86 and recall was 0.79, suggesting that while most predicted fatigue cases were correct, some actual fatigue instances were missing. These values reflect reasonable precision for both classes but reveal a performance gap in fatigue recall, where several fatigue cases misclassified as non-fatigue. This impacts the F1 score and suggests the model is conservative in identifying fatigue under noisy conditions. The confusion matrix, presented in Fig. 16, provides further insight into model behavior on the test set. The model correctly identified 19 fatigue cases as true positives and 15 non-fatigue cases as true negatives. It misclassified 3 non-fatigue instances as fatigue (false positives) and missed 5 actual fatigue cases (false negatives).

The presence of five false negatives is particularly notable. In the context of driver monitoring, such errors are critical, as undetected fatigue events undermine the utility of the system. The reduced recall for the fatigue class (0.79) reflects this limitation and is influenced by subject-level label imbalance and variability in physiological fatigue signatures. The results from the MAUS dataset demonstrate that ECG-derived statistical features, when processed through the proposed classification pipeline, can distinguish between fatigue and non-fatigue states under cognitively demanding conditions. The model achieved test accuracy of 80.95 % and an F1 score of 82.61 %, with balanced precision across classes. However, drop in fatigue recall and the number of false negatives emphasize the challenges of fatigue detection in high-variability datasets.

Table 8
Summary of heart rate datasets used for fatigue detection. This table outlines participant count, total duration, and type of recording environment for each dataset.

Dataset	Subjects	Hours / Trials	Avg. Samples	Environment
OpenDriver	18	~16 h	~2000+	Real-world driving
DriveDB	16	Multiple runs	~300–500	Lab simulation
MAUS	20+	6 tasks/subject	~600–800	Cognitive tests

6.4. Comparative analysis across heart datasets

6.4.1. Overview and objective

To validate the reliability and scalability of the proposed ECG-based fatigue detection system, we conducted a structured comparative analysis across three publicly available heart signal datasets: OpenDriver, DriveDB, and MAUS. These datasets vary significantly in experimental setting, subject diversity, and the nature of fatigue they capture [44]. By applying a unified preprocessing and classification pipeline across them, we assessed how well the model generalizes across both physical and cognitive fatigue contexts. This comparison not only benchmarks accuracy but also analyzes misclassification tendencies, feature relevance, and robustness in diverse data environments.

6.4.2. Statistical analyses

Each dataset was first examined for its size, duration, and acquisition environment. This initial profiling helps contextualize performance outcomes and highlights differences in experimental rigor and fatigue induction mechanisms. The summary is enlisted in Table 8.

The evaluation across the OpenDriver, DriveDB, and MAUS datasets provides a comprehensive view of the model's generalization capabilities across diverse fatigue scenarios. OpenDriver offers the most naturalistic and extensive dataset (~16 h, 18 subjects), DriveDB introduces stress-focused trials in controlled settings, and MAUS captures mental fatigue under cognitive tasks. This variety ensures that the model is evaluated against both real-world variability and structured experimental conditions. As shown in Table 1, these datasets differ significantly in context and scale. OpenDriver's size and ecological validity contributed to its robust performance, with a mean cross-validation accuracy of 97.6 % and test accuracy of 96.4 % (Table 2), indicating excellent generalization. DriveDB followed with 90 % CV accuracy but achieved 100 % on the test set, suggesting overfitting due to its smaller sample size. MAUS recorded the lowest CV (70 %) and test accuracy, reflecting the difficulty of detecting mental fatigue and higher subject variability. Overall, the findings underscore that ECG-based fatigue detection performs best in naturalistic settings, though mental fatigue detection remains more complex and less reliable without additional modeling. We compared the model's classification performance based on cross-validation accuracy and test accuracy to assess its consistency and generalization capacity.

OpenDriver yielded both high cross-validation and test accuracy. Its natural driving data provided a rich set of fatigue variations, allowing the model to learn discriminative patterns effectively. The small gap between training and test performance also indicates strong generalization. DriveDB achieved perfect test accuracy (100 %), but this is misleading due to the small test set (only 6 samples). Moreover, a 90 % CV accuracy with high variance suggests potential overfitting on specific runs. MAUS presented the lowest CV and test accuracy. This possibly attributed to variability in cognitive fatigue expression, less physiological consistency, and label noise due to subjectively defined mental fatigue thresholds. Fig. 17 demonstrates consistent model performance across cross-validation folds (blue), highlighting the OpenDriver dataset's strong generalization capability during training whereas standard deviation of cross-validation accuracies can be visualized (red). OpenDriver exhibits minimal variability (below 1 %), underscoring the reliability and stability of its results. In contrast, DriveDB shows a markedly

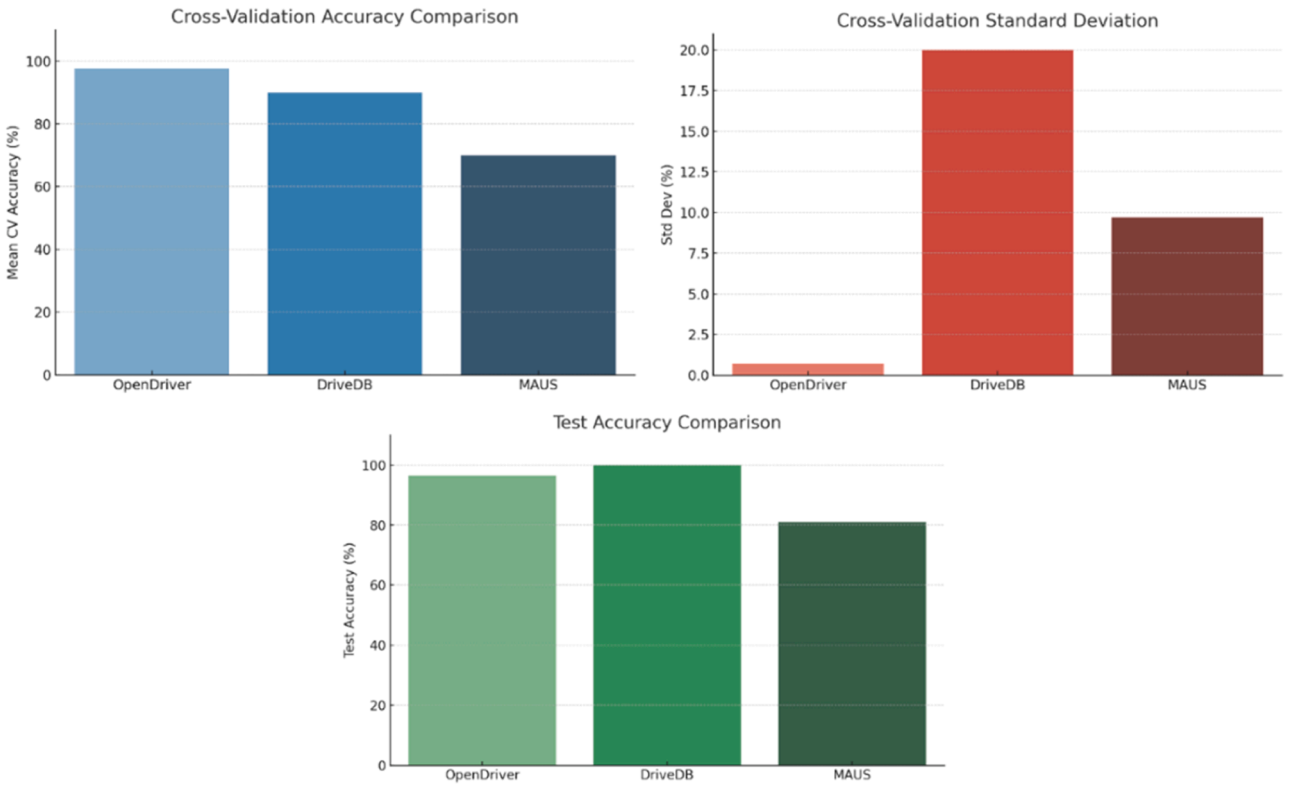


Fig. 17. Cross-validation accuracy showing model stability during training (blue), cross validation standard deviation (red), comparison of test set accuracy (green) across all datasets.

Table 9
Cross-validation accuracy, test set accuracy and FI score for each dataset.

Dataset	Mean CV Accuracy	Test Accuracy	F1 Score
OpenDriver	97.6 %	96.4 %	95.8 %
DriveDB	90.0 %	100 %	100 %
MAUS	70.0 %	80.9 %	82.6 %

higher variance of around 20 %, indicating a substantial risk of overfitting, likely due to its limited sample size. The MAUS dataset displays moderate variance, reflecting the challenges of modeling mental fatigue amid cognitively demanding conditions and inter-subject variability.

The F1 score listed in Table 9, offers a balance between precision (avoiding false positives) and recall (avoiding false negatives). This is especially important in detecting driver fatigue, where both missed detections and false alarms have serious safety implications. OpenDriver exhibited a strong F1 score, indicating balanced performance in detecting both fatigued and alert states. This further supports the pipeline's suitability for real-world deployment. DriveDB's perfect F1 score again reflects its small test set. Although all six test samples were correctly classified, the limited size means the F1 is not a statistically robust indicator. MAUS revealed the lowest F1 score due to elevated false negatives. Mental fatigue tends to manifest in more subtle ECG changes than physical fatigue, making it harder for the model to detect.

Fig. 18 (left) illustrates the test F1 scores achieved on the OpenDriver, DriveDB, and MAUS datasets. The F1 score balances precision and recall, offering a more comprehensive view of classification performance, particularly in datasets with potential class imbalance. DriveDB demonstrates a perfect F1 score (100 %), reflecting ideal model balance under controlled conditions. OpenDriver also performs strongly (~96 %), suggesting effective generalization in real-world driving scenarios. In contrast, the MAUS dataset yields a lower F1 score (~83 %), due to the increased variability and complexity of mental fatigue induced by cognitive workload tasks. The comparative trend of test

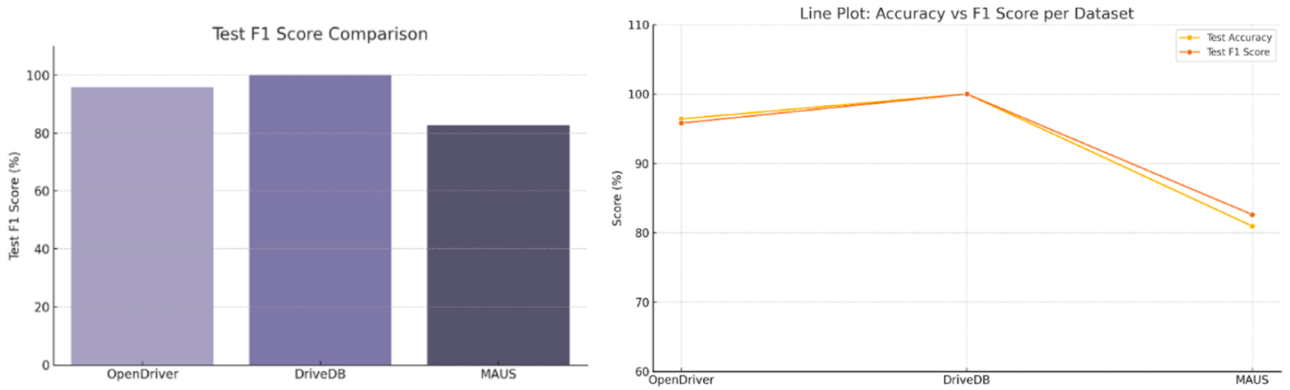


Fig. 18. F1 score comparison reflecting model balance between precision and recall (left) and, line plot showing both accuracy and F1 trend across datasets (right).

Table 10
Raw confusion matrix components (TP, TN, FP, FN) for each dataset's test set.

Dataset	TP	TN	FP	FN
OpenDriver	115	153	6	4
DriveDB	4	2	0	0
MAUS	19	15	3	5

accuracy and F1 scores across the three datasets is shown in Fig. 18 (right). The close alignment between both metrics indicates that the model maintains balanced precision and recall, without overfitting to one metric. OpenDriver and DriveDB both exhibit high performance, with DriveDB reaching 100 % in both metrics. However, the drop observed for the MAUS dataset in both accuracy and F1 score highlights the model's reduced robustness when applied to cognitively complex and variable fatigue conditions. This trend reinforces the challenge of detecting mental fatigue compared to physical fatigue.

The confusion matrix provides insight into specific misclassification trends, such as whether the model struggles more with false positives or false negatives. The comparison of the confusion matrix component is enlisted in Table 10. OpenDriver achieved a near-perfect balance between true positives and true negatives, with very few false outcomes. This reflects the maturity and quality of the dataset. DriveDB, though showing perfect classification, is based on a very small test group. The absence of false results here should not be overinterpreted. MAUS showed a tendency toward false negatives, a serious issue for fatigue detection systems, as missing a fatigued driver is more dangerous than raising a false alarm.

Fig. 19 on the left represents the stacked bar chart giving a consolidated view of model performance by visualizing both test accuracy and F1 score across the OpenDriver, DriveDB, and MAUS datasets. The DriveDB dataset achieves the highest combined performance, reflecting perfect classification results under controlled simulation. OpenDriver follows closely, maintaining strong consistency and generalizability in real-world driving conditions. MAUS records the lowest cumulative score, reinforcing the challenge of detecting cognitive fatigue. This visualization offers an intuitive performance comparison by integrating two key evaluation metrics into a unified format. Fig. 19 (right) presents the raw classification outcomes, true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), for each dataset. The OpenDriver and DriveDB datasets demonstrate high counts of TP and TN, with minimal misclassifications, underscoring strong predictive performance (Bosman et al., 2013). However, MAUS exhibits a higher number of false negatives, indicating a tendency of the model to miss fatigue instances in cognitively demanding scenarios. This breakdown highlights the importance of context, specific model tuning, especially when dealing with mental workload variability.

6.4.3. *Feature importance comparison*

Identifying which features drive classification helps interpret model

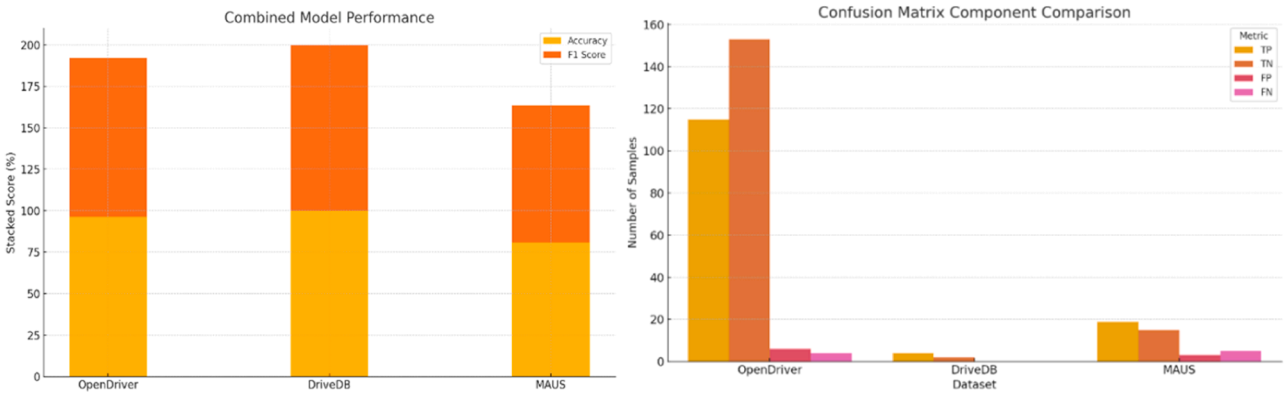


Fig. 19. Stacked bar chart combining test accuracy and F1 score (left), Raw count comparison of TP, TN, FP, and FN (right) for each dataset.

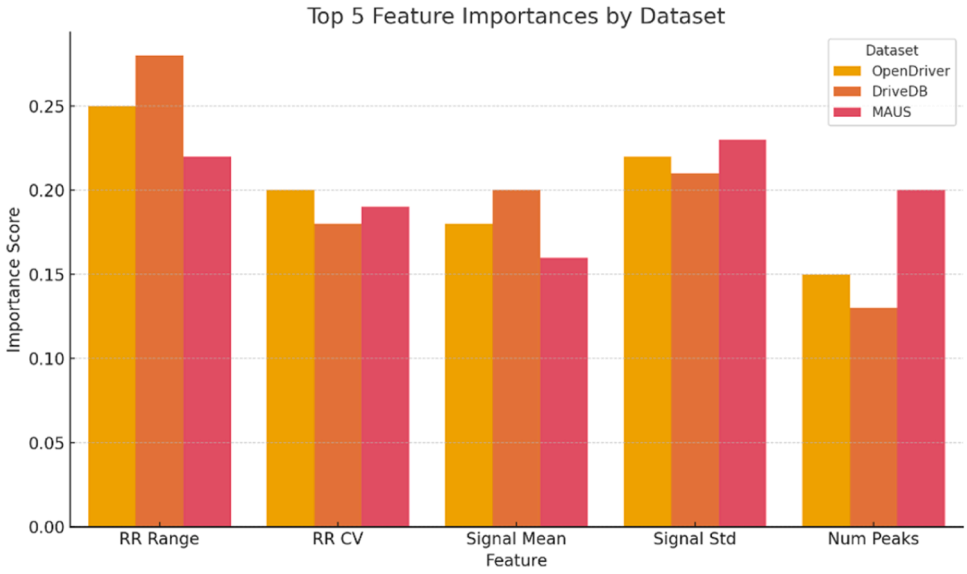


Fig. 20. Top 5 feature importance's per dataset from Random Forest.

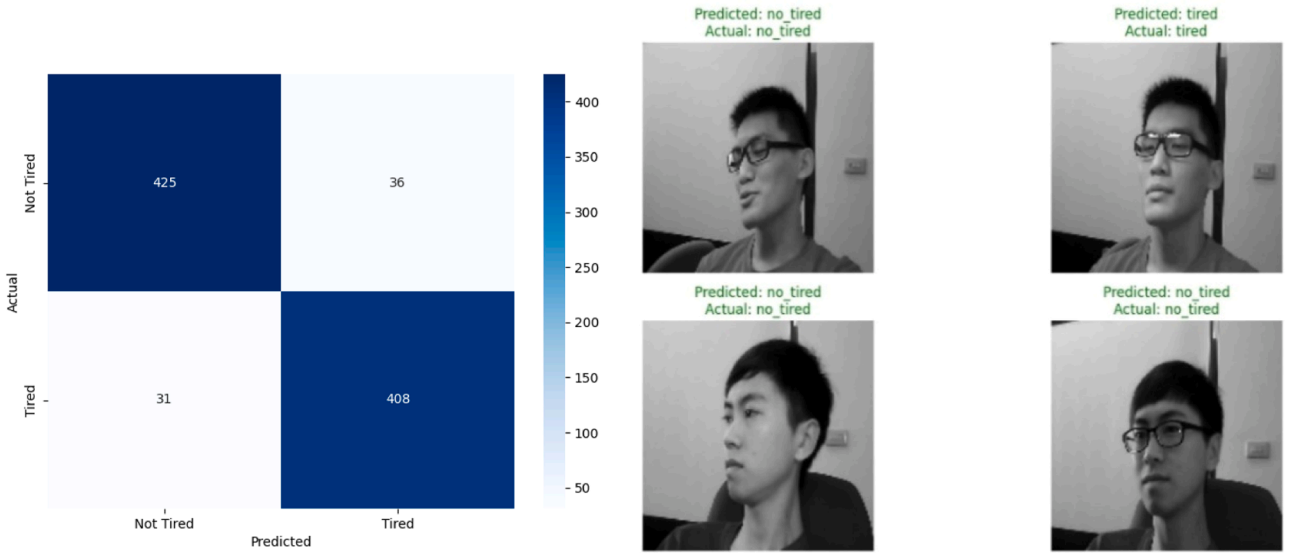


Fig. 21. Confusion matrix of CNN predictions on the YawDD test set (left), and examples of CNN predictions on the YawDD test set (right).

behavior and adapt the system for different use cases. A comparison of features is represented in Fig. 20. In OpenDriver, RR Range and Signal Mean were the dominant features. These reflect heart variability and average amplitude, strong indicators of physical fatigue. In DriveDB, the same features were effective, due to short but intense periods of induced fatigue. In MAUS, Signal Std and Number of Peaks rose in importance. These features capture irregularities and local fluctuations in the ECG signal, hallmarks of cognitive load and attention shift during mental tasks. This variability in feature importance underscores that several types of fatigue (physical vs. mental) affect ECG signals in distinct ways. A flexible or hybrid model may be needed to manage both equally well.

6.4.4. *Insights*

This comparative evaluation reinforces the strength of our fatigue detection pipeline in real-world physical fatigue scenarios and highlights the challenges in generalizing it to mental fatigue conditions. Key takeaways depicted that OpenDriver provided the most reliable and generalizable results, with balanced metrics and low variance. It is well-suited for deployment in real-world driving applications. DriveDB appeared to perform perfectly but must be interpreted with caution due to its small test size. The high CV variance also indicates sensitivity to subject/session differences. MAUS, while valuable for studying cognitive fatigue, revealed limitations in current ECG-only detection methods. The high false negative rate suggests a need for enhanced modeling strategies, such as multimodal fusion (e.g., EEG, eye tracking) or temporal modeling (e.g., LSTM). In conclusion, our rule-based feature extraction combined with Random Forest classification is effective for physical fatigue detection, but further optimization is necessary to achieve robust performance across all forms of driver fatigue.

6.5. *Eye fatigue detection using the YawDD dataset*

Upon evaluation using the held-out test set of 870 samples, CNN achieved an impressive accuracy of 93 %. Other performance metrics, including precision, recall, and F1-score, were all recorded at 93 %. The classification report indicated a precision of 0.93 and recall of 0.92 for the “not tired” class, while the “tired” class achieved a precision of 0.92 and a recall of 0.93. These values confirm that the model maintained balanced performance across both categories. The detailed metrics are as follows: accuracy was 93 %, precision was 93 %, recall was 93 %, and the F1 score was also 93 %.

6.5.1. *Classification report, confusion matrix and visual prediction results*

For the Not Tired class, the model achieved a precision of 0.93 and a recall of 0.92, indicating a strong ability to correctly identify non-fatigued instances with minimal false positives. For the Tired class, the precision was 0.92 and the recall was 0.93, demonstrating effective detection of fatigued samples with few false negatives. These class-wise results reflect the model’s balanced and reliable performance across both categories.

The confusion matrix presented in Fig. 21 (left) shows that the system correctly identified 408 tired cases (true positives) and 425 non-tired cases (true negatives), while misclassifying 36 instances as tired (false positives) and 31 as not tired (false negatives). This corresponds to an overall error rate of 7.8 %, demonstrating high classification accuracy with minimal misclassifications. A qualitative analysis of prediction accuracy conducted using test samples, illustrated in Fig. 21 (right). Correct classifications are shown with blue boxes, while misclassified images were enclosed in red. Most misclassifications occurred in frames with ambiguous visual states, such as partially closed eyes or poor lighting conditions.

Correct classifications indicated by green-bordered boxes, while misclassifications are shown with red-bordered boxes. The misclassifications primarily caused by ambiguous visual states, such as half-open eyes or poor lighting conditions, which made it challenging for the model to distinguish between tired and not tired states accurately.

6.6. *Eye fatigue detection using the TinyML eye dataset*

To further evaluate the generalizability of our CNN model, we employed the TinyML Eye Dataset, which comprises a total of 2900 tightly cropped grayscale eye-region images. These images captured under highly controlled environmental conditions, including consistent lighting and fixed camera angles, thereby minimizing background noise, and enhancing the clarity of fatigue-relevant features. Originally, the dataset included four distinct class labels; however, for the purpose of aligning with our binary classification objective, the labels consolidated into two primary categories: “tired” and “not tired.” This relabeling ensured consistency with the model’s fatigue detection framework.

Each image underwent a standardized preprocessing pipeline to ensure uniformity and compatibility with the CNN architecture. Specifically, all images were resized to a resolution of 150 $\times$ 150 pixels to balance spatial detail with computational efficiency. They were subsequently converted to grayscale to simplify the input space and

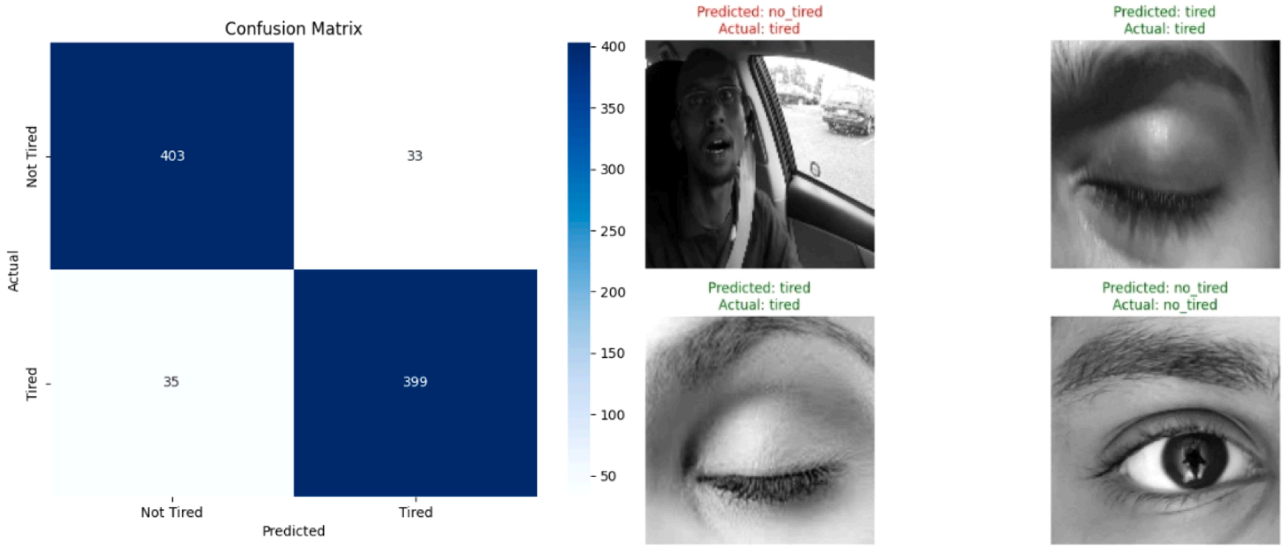


Fig. 22. Confusion matrix of CNN predictions (left), Sample test set predictions (right) on TinyML Eye Dataset.

normalized to ensure consistent pixel intensity distributions. The dataset was then stratified and split into two subsets: 2030 images were allocated for training, while the remaining 870 images used exclusively for testing. This division allowed for robust training while preserving a sizable held-out set to evaluate model generalization. By applying this preprocessing and restructuring protocol, the TinyML Eye Dataset served as an effective benchmark to assess the CNN model's ability to detect visual indicators of fatigue under ideal and noise-free conditions. The carefully controlled image acquisition environment, characterized by stable lighting and consistent angles, further minimized the potential confounding factors often present in real-world scenarios, allowing us to isolate and evaluate the model's core discriminatory capabilities. The same convolutional neural network (CNN) architecture previously employed for the YawDD dataset applied to the TinyML Eye Dataset to maintain consistency in evaluation. The model trained for 20 epochs using the Adam optimizer and categorical cross-entropy as the loss function, reflecting a standard setup for multi-class (later adapted to binary) image classification tasks. Training performance showed a consistent upward trend: accuracy at the first epoch was recorded at 55.5 %, indicating the model's initial attempts at learning from the data. By the 20th epoch, training accuracy had significantly improved to 95.7 %, highlighting the model's capacity to learn discriminative patterns from the tightly controlled grayscale eye images. Additionally, the final training loss was measured at 0.1198, a relatively low value that suggests the network achieved stable convergence with minimal overfitting. This progression reflects architecture's effectiveness when applied to structured visual data, further supporting its suitability for fatigue detection tasks.

6.6.1. Report breakdown

On the test set, the model achieved a classification accuracy of 92 %, indicating a strong overall performance in distinguishing between tired and not tired states. In addition to accuracy, other key evaluation metrics, precision, recall, and F1-score, were each recorded at 92 %, reflecting a well-balanced classifier. These consistent values suggest that the CNN model-maintained symmetry in its ability to both correctly identify tired individuals and avoid misclassifying those who were not tired. A detailed classification report further reinforced this observation. For the "not tired" class, the model achieved a precision of 0.92 and a recall of 0.92, showing that it reliably identified non-fatigued eye images with minimal error. The "tired" class also registered a precision of 0.92 and a recall of 0.92, confirming that the model was equally adept at detecting signs of fatigue. This balance across both classes is particularly important in safety-critical applications, as it minimizes the risk of both false positives and false negatives.

Table 11
Comparison of YawDD and TinylML.

Metric	YawDD	TinyML
Accuracy	93 %	92 %
Precision	93 %	92 %
Recall	93 %	92 %
F1 Score	93 %	92 %

6.6.2. Confusion matrix and visual output samples

The confusion matrix, shown in Fig. 22 (left), indicates that the CNN accurately predicted 399 tired cases, representing the true positives (TP), and 403 not tired cases, representing the true negatives (TN). However, the model also misclassified 33 instances as tired when they were not (false positives), and 35 tired instances incorrectly labeled as not tired (false negatives). These misclassifications resulted in a total error rate of 7.8 %, which, while modest, highlights areas for further refinement in borderline visual cases. Fig. 22 (left) visualizes this performance distribution, providing insight into the model's behavior on the TinyML Eye Dataset. Visual predictions on test data displayed in Fig. 22 (right), where correctly predicted samples are labeled in green, and incorrect ones in red. Most errors linked to low-contrast eye images or those exhibiting visually borderline states between alertness and fatigue. The controlled and noise-free environment in which the TinyML dataset was captured contributed to strong classification accuracy. However, due to its homogeneous conditions, the model may struggle when exposed to the wider variability present in real-world settings.

For deployment in naturalistic driving scenarios, integrating this dataset with more varied data like YawDD is recommended (Saseendran et al., 2019). The model consistently predicted eye states that were clearly open or shut with high accuracy. However, most of the classification errors occurred in cases involving low-contrast images or borderline eye states, such as partially open eyes. These ambiguous visual conditions reduced the model's confidence and contributed to misclassifications.

6.7. Comparative evaluation of eye-based CNN models

To compare the effectiveness of the CNN-based models trained on both datasets, we conducted a side-by-side evaluation using key

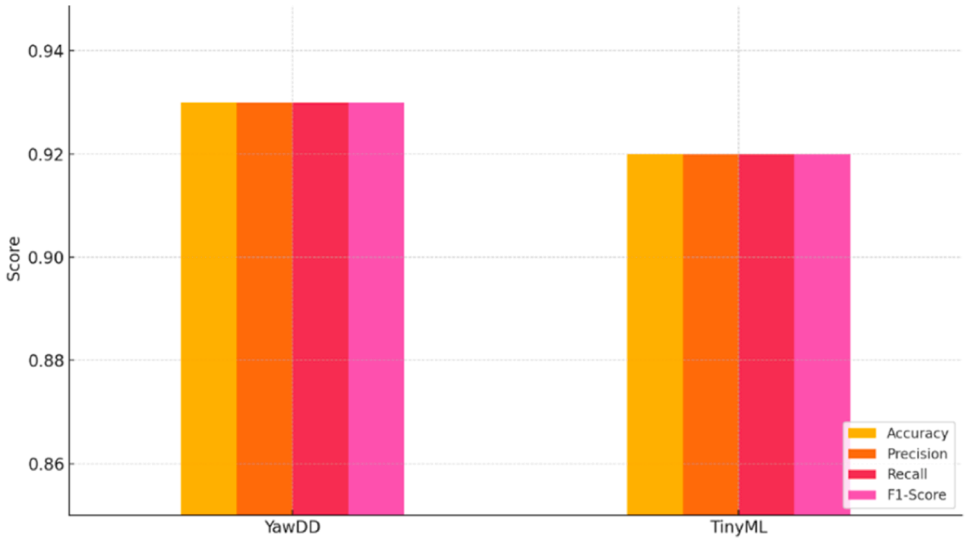


Fig. 23. Bar chart comparing CNN performance metrics for both eye datasets.

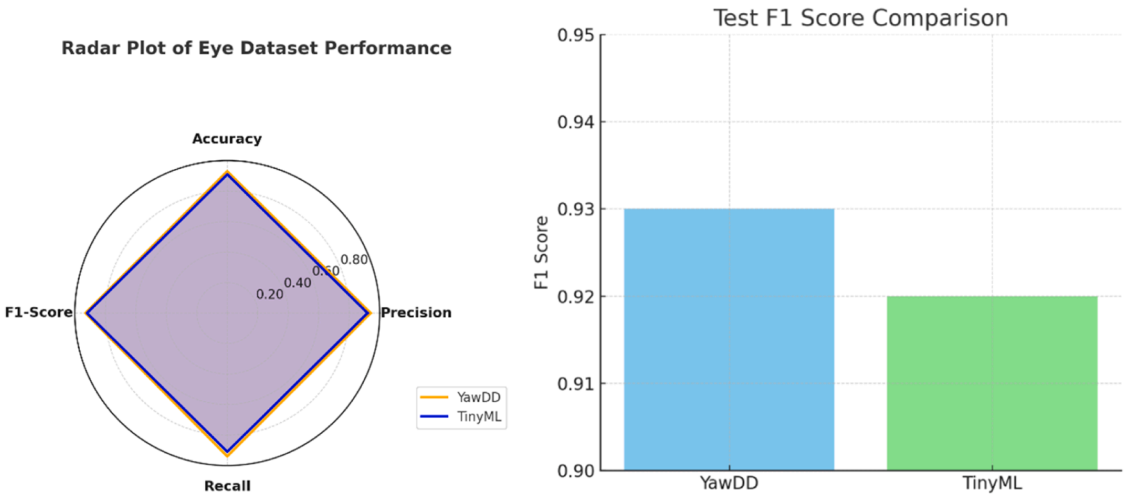


Fig. 24. Radar plot showing the overall profile of model performance (left), Test F1 score comparison between YawDD and TinyML eye datasets (right).

performance metrics. Specifically, the YawDD model achieved 93 % accuracy, precision, recall, and F1-score, while the TinyML model maintained 92 % across all four metrics (Table 11). This 1 % lead, though marginal, suggests better robustness of the YawDD model under diverse conditions.

Both models performed exceptionally well, with YawDD showing a

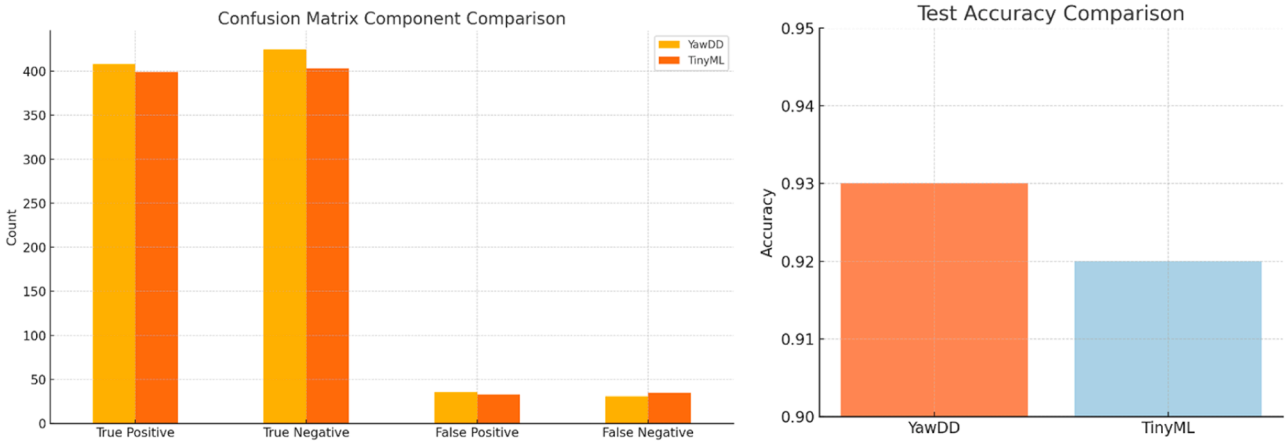


Fig. 25. Confusion matrix component comparison between the two datasets (left), Test accuracy comparison for YawDD and TinyML eye fatigue classifiers (right).

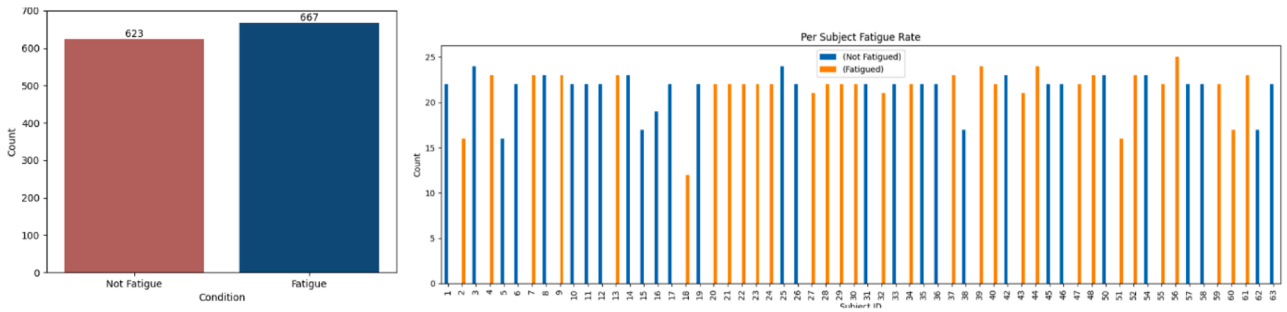


Fig. 26. Distribution of fatigue and non-fatigue labels in the breathing dataset (left), fatigue vs. non-fatigue sample count per driver (right).

consistent 1 % lead. The bar chart confirms that YawDD's natural variability improved generalization. A bar chart shown in Fig. 23 confirms this trend, visualizing the close performance alignment of both models with YawDD consistently maintaining a slight advantage. In Fig. 24, a radar plot further reveals that the YawDD model forms a wider and more symmetrical shape across the metric dimensions, while the TinyML plot appears slightly narrower, indicating reduced consistency across classes.

6.7.1. Statistical analyses

The radar plot provides a visual summary of model performance across key evaluation metrics, accuracy, precision, recall, and F1 score, for both datasets. The YawDD model forms a wider and more symmetrical polygon on the radar chart, reflecting balanced and consistently high performance across all metrics (Fig. 24). This suggests that the model trained on YawDD is more robust and generalizable. In contrast, the polygon representing the TinyML model is slightly narrower, indicating marginally lower scores in certain metrics and thus slightly reduced robustness overall. The test F1 score comparison in Fig. 24 (right) reveals a slight performance advantage for the YawDD dataset, with an F1 score of 0.93 compared to 0.92 for the TinyML dataset. This marginal difference suggests that while both models performed exceptionally well, the model trained on YawDD benefitted from the natural variability present in real-world driving conditions, potentially enhancing its generalization capability.

YawDD had more correct predictions (TP, TN) and fewer errors. TinyML showed slightly higher FP and FN, which could affect real-world deployment. In terms of F1-score alone, the YawDD model scored 0.93, whereas the TinyML model followed closely at 0.92. This comparison highlighted in Fig. 25 (left). Analysis of the confusion matrix components, shown in Fig. 25, revealed that the YawDD model produced higher counts of true positives and true negatives, while the TinyML model exhibited slightly more false positives and false negatives, factors that could impact safety in real-world fatigue detection systems. The test set accuracy comparison in Fig. 25 (right) highlights a minor performance difference between the two CNN models.

The classifier trained on the YawDD dataset achieved an accuracy of 93 %, while the model trained on the TinyML Eye dataset reached 92 %. This 1 % advantage reflects YawDD's richer variability and naturalistic eye-state representations, which may have contributed to slightly better generalization and robustness in real-world conditions. Both models exhibit tight alignment between accuracy and F1-score, suggesting strong calibration and balanced predictive performance. This consistency indicates that neither model favors precision over recall, nor can both generalize well without significant bias toward either class. However, the YawDD model's trajectory is slightly more stable, suggesting improved reliability across various evaluation metrics. Test accuracy alone reaffirms the marginal performance gap, with YawDD achieving 93 % and TinyML reaching 92 %.

6.7.2. Insights

Both CNN-based classifiers demonstrated strong fatigue detection capabilities with high accuracy and balanced classification. The YawDD dataset, collected under real-world driving conditions, provided the benefit of broader variation and thus stronger generalization, making it better suited for deployment scenarios. In contrast, the TinyML dataset, while highly effective in controlled environments, may not fully reflect the unpredictable nature of real-life fatigue indicators. A hybrid training strategy combining both datasets could yield even more resilient and adaptable models, suitable for dynamic in-vehicle fatigue monitoring systems.

Table 12
Top 10 most informative features of feature section analysis and correlation.

Rank	Feature Name	Correlation
1	RSA_Gates_SD_Bl	0.310
2	HRV_C1d_Bl	0.305
3	HRV_C2a_Bl	0.266
4	HRV_DFA_alpha1_ExpRange_Bl	0.263
5	RSA_P2T_SD_Bl	0.246
6	HRV_AI_Bl	0.242
7	RRV_SD2SD1_Bl	0.231
8	HRV_Ca_Bl	0.218
9	HRV_GI_Bl	0.215
10	RSP_Amplitude_Mean_Bl	0.215

6.8. Fatigue detection using breathing-based features

Following preprocessing and subject-wise data splitting, the model was trained and evaluated on the respiration and HRV-based dataset to assess its effectiveness in fatigue classification. The preserved subject independence ensured that evaluation metrics reflected true generalization across individuals.

6.8.1. Statistical analyses

An initial exploratory analysis revealed a slight imbalance in the number of fatigue and non-fatigue samples. Specifically, 667 samples labeled as fatigued and 623 as non-fatigued, as illustrated in Fig. 26 (left), which shows the overall label distribution. To better understand inter-subject diversity, the dataset also analyzed per individual. Fig. 26 (right) displays the count of fatigue vs. non-fatigue samples across all subjects. While the sample count per subject varied, the distribution between classes remained consistent, supporting the model's generalization across different individuals.

The dataset included a wide range of biosignal-derived features. To reduce dimensionality and improve model performance, a Pearson correlation analysis (Cleophas & Zwinderman, 2018) was conducted to identify the most relevant predictors. Based on this analysis, the top 10 features with the strongest correlation to the fatigue label were selected for training. The most informative feature was RSA_Gates_SD_Bl (a measure of respiratory sinus arrhythmia variability) with a correlation coefficient of 0.31, followed by other features. The complete ranking is shown in Table 12, which lists each selected feature along with its correlation value. These features were used as input in the machine

Table 13
Classification report for fatigue prediction using breathing data (Random Forest).

Class	Precision	Recall	F1-score	Support
Not Fatigue (0)	0.35	1.00	0.51	112
Fatigue (1)	1.00	0.24	0.39	279
Accuracy	-	-	0.46	391

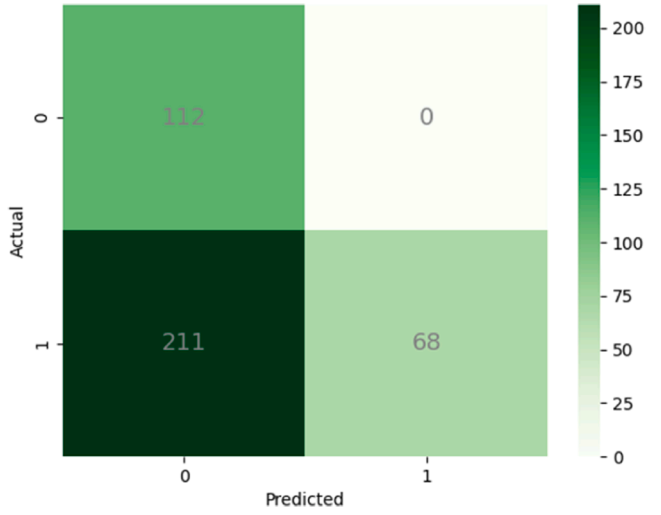


Fig. 27. Confusion matrix of random forest classifier on breathing-based dataset.

learning model and were critical in defining the classifier's decision boundaries.

A Random Forest classifier trained using the selected features, with the model configured to use 42 estimators and balanced class weights to address the slight imbalance between fatigue and non-fatigue labels. To assess the stability and generalizability of the model, a 5-fold cross-validation was conducted. The resulting fold accuracies were 0.528, 0.883, 0.756, 0.494, and 0.877, respectively. The mean cross-validation accuracy across all folds calculated as 70.76 %, with a standard deviation of 16.71 %. This substantial variation in performance across folds suggests that the model's effectiveness was highly dependent on the training subset used in each iteration. Such variability could stem from differences in ECG signal quality or inter-subject heterogeneity within the dataset, raising concerns about the potential overfitting of the model to certain individuals or data segments. The model's performance evaluated on a hold-out test set comprising 30 % of the total data, stratified by subject to maintain label balance. The results demonstrated a significant performance drop compared to the cross-validation scores, with an overall test accuracy of only 46.0 %. The full classification report is summarized in Table 13.

The model exhibited an unusual performance pattern. For the non-fatigue class (label 0), it achieved a precision of 0.35, a perfect recall of 1.00, and an F1-score of 0.51. In contrast, for the fatigue class (label 1), the model recorded a precision of 1.00 but a much lower recall of 0.24, resulting in an F1-score of 0.39. These results suggest a significant imbalance in the model's predictive behavior. Specifically, the model displayed a strong bias toward classifying instances as non-fatigue, driven by overlapping feature distributions between classes or an underlying model bias. This skewed behavior undermines the classifier's effectiveness in correctly identifying fatigue cases.

The confusion matrix presented in Fig. 27 illustrates the prediction breakdown in detail. The model correctly classified 68 fatigue instances (true positives) and 112 non-fatigue instances (true negatives). Notably, there were no false positives, meaning no non-fatigue cases incorrectly labeled as fatigue. However, the model misclassified 211 fatigue cases as non-fatigue (false negatives), resulting in an extremely high false negative rate. This is particularly concerning in the context of fatigue detection, where missing actual fatigue cases poses serious risks. In critical applications like driving or industrial monitoring, such misclassifications could lead to hazardous situations, emphasizing urgent need for a balanced and sensitive classification approach.

Table 14
Ablation study.

Configuration	Accuracy (%)	F1-Score (%)
ECG only	96.4	95.8
Eye only	93.0	93.0
Respiration only	46.0	43.0
ECG + Eye	97.8	96.9
All modalities	98.1	97.2

6.8.2. *Discussion and insights*

Despite using meaningful physiological features and applying careful subject-wise validation, the Random Forest model's performance was significantly inferior to that of the ECG and eye-based CNN modules. Several factors contributed to this underperformance, like weak or non-linear relationships between the selected features and the fatigue label, as indicated by low correlation coefficients. Natural variability in breathing patterns, even among non-fatigue individuals, leading to false positives or ambiguous signals. Evidence of overfitting, particularly during cross-validation, is seen in the large variance in fold accuracy scores. These limitations make the current model unsuitable for standalone deployment in fatigue detection. However, breathing data may still hold value as part of a multi-modal fusion framework, where it could support more reliable signals (such as ECG or eye tracking) and help correct their potential misclassifications. To improve performance in future iterations, enhancements are recommended. First, advanced feature engineering techniques, including frequency-domain analysis or time-series transformations, could reveal hidden patterns in the breathing signals. Additionally, ensemble modeling techniques such as stacking or boosting may help overcome the limitations of individual classifiers. Temporal modeling approaches like recurrent neural networks (RNNs) or long short-term memory (LSTM) networks could be particularly effective in capturing breathing rhythms and transitions over time. Addressing class imbalance through rebalancing strategies such as Synthetic Minority Over-sampling Technique (SMOTE) may also improve sensitivity to the class of fatigue. Finally, integrating breathing data with other biosignals in a multi-modal architecture could yield a more holistic and robust fatigue profiling system.

6.9. *Ablation study*

To evaluate the individual and combined contributions of different sensing modalities, an ablation study was conducted by systematically training and evaluating models using unimodal and multimodal configurations. As summarized in Table 14, the ECG-based model achieved strong standalone performance, confirming the reliability of heart rate variability features for fatigue discrimination. The eye-based model also demonstrated competitive accuracy, highlighting the effectiveness of visual fatigue cues. In contrast, the respiration-based model exhibited substantially lower performance, indicating limited discriminative capability when used in isolation. Combining ECG and eye modalities resulted in a notable improvement over unimodal configurations, suggesting complementary information between physiological and visual signals. The full multimodal configuration achieved the highest overall performance; however, the marginal improvement over the ECG-eye combination indicates that respiration contributes only limited additional benefits in the current setup. These findings demonstrate that multimodal fusion enhances robustness primarily through ECG and

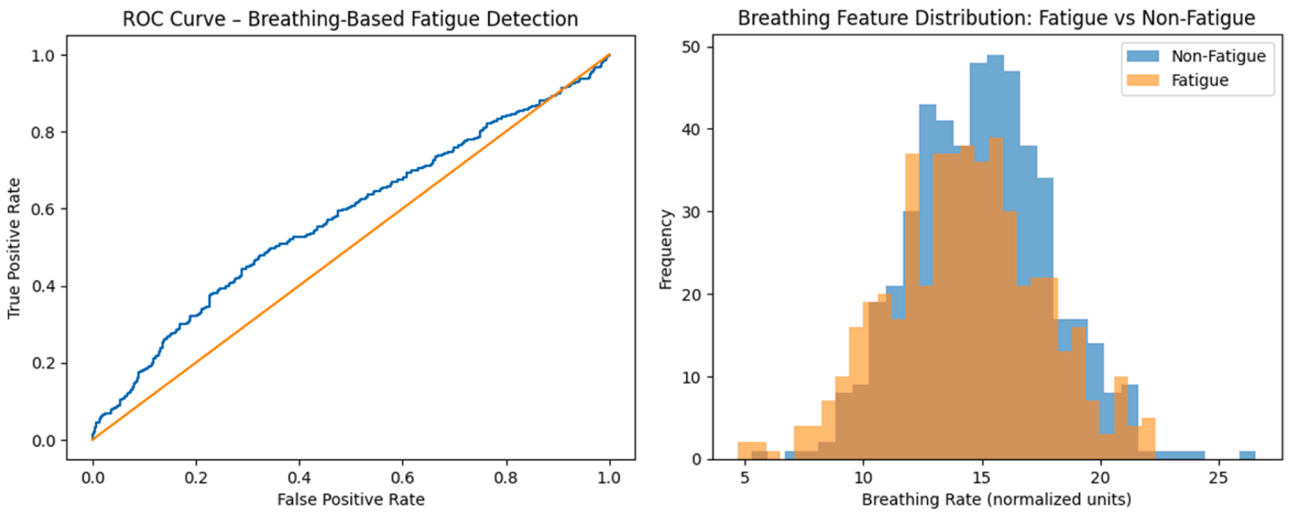


Fig. 28. ROC curve for breathing-based fatigue detection (right), Distribution of breathing features under fatigue and non-fatigue conditions (left).

vision, while respiration functions as a secondary supportive modality rather than a dominant contributor.

6.10. *Comparative analysis of fatigue detection modalities*

Fatigue detection can be approached through various physiological and behavioral modalities, each offering unique advantages and limitations. This section presents a comparative analysis of three such modalities investigated in this study: breathing-based physiological features, ECG-derived heart signals, and eye-based image classification. The evaluation encompasses multiple performance indicators, including accuracy, precision, recall, F1-score, cross-validation (CV) stability, confusion matrix breakdowns, and feature interpretability. The goal is to critically assess the trade-offs of each method and inform selection for real-world fatigue monitoring systems. The respiration-based fatigue detection model exhibited substantially weaker performance compared to ECG- and eye-based modalities, primarily due to significant overlaps in feature distributions between fatigue and non-fatigue classes. As illustrated by the confusion matrix and receiver operating characteristic analysis, the model demonstrated high precision but poor recall, resulting in an excessive number of false negatives. This behavior indicates that while the classifier was conservative in predicting fatigue, it frequently failed to detect true fatigue instances. Feature distribution analysis reveals limited class separability for respiration-derived features, suggesting that breathing patterns alone may not capture sufficiently distinctive fatigue signatures under heterogeneous driving and cognitive conditions. Consequently, respiration signals are better interpreted as a complementary modality that may enhance robustness when fused with more discriminative signals, rather than as a reliable standalone indicator of fatigue.

Receiver Operating Characteristic (ROC) curve of the Random Forest classifier trained on respiration-based features from the Driver Respiration Dataset is provided in Fig. 28 (left). The shallow curve and low AUC indicate limited discriminative capability of breathing features when used as a standalone modality. The respiration-based classifier achieved a low area under the ROC curve (AUC ≈ 0.57), indicating weak separability between fatigue and non-fatigue states and performance close to random classification. Histogram showing the distribution of a representative breathing feature (normalized breathing rate) for fatigue and non-fatigue classes is shown in Fig. 28 (right). Significant overlap between the two distributions highlights the limited discriminative power of respiration-based features. As shown in Fig. 28 (right), breathing features exhibit substantial overlap between fatigue and non-fatigue conditions, explaining the respiration model's reduced classification performance and elevated false-negative rate.

Table 15
Test performance metrics.

Modality	Accuracy	Precision	Recall	F1-Score
Eye (YawDD)	93 %	93 %	93 %	93 %
Eye (TinyML)	92 %	92 %	92 %	92 %
Heart (OpenDriver)	96.4 %	95 %	97 %	95.8 %
Heart (MAUS)	80.9 %	83 %	82 %	82.6 %
Breathing	46.0 %	81 %	46 %	43 %

6.10.1. *Comparative evaluation metrics across modalities*

To establish a quantitative baseline, we first compared test performance metrics across the three modalities, focusing on their ability to generalize to unseen subjects. As reported in Table 15, eye-based models achieved robust and balanced results, with the YawDD and TinyML datasets yielding accuracies of 93 % and 92 %, respectively. Heart-based models, particularly the one trained on the OpenDriver dataset, demonstrated superior precision and recall, achieving a test accuracy of 96.4 % and an F1-score of 95.8 %. The MAUS-based heart model, while slightly lower in performance, still maintained a strong F1-score of 82.6 %.

In contrast, the breathing-based model lagged significantly behind, with a test accuracy of only 46 % and an F1-score of 43 %. Despite high precision (81 %), its poor recall (46 %) indicates that it frequently failed to detect fatigue, a critical limitation for safety applications. This metric imbalance is a key concern, especially in systems where missing fatigued states can have severe consequences. A deeper evaluation of metric consistency is illustrated in Fig. 29, which plots accuracy and F1-score across all modalities.

Ideally, a well-performing classifier should exhibit minimal divergence between these two metrics. The eye-based and heart-based models demonstrate such alignment, reflecting balanced precision and recall. However, the breathing-based model exhibits a pronounced gap, with accuracy remaining moderate while F1-score declines steeply. This divergence underscores the model's inadequate performance in capturing positive fatigue instances, driven primarily by the high false negative rate while breathing diverges, confirming recall imbalance. To understand the nature of misclassifications, the confusion matrix components analyzed. Fig. 29 presents the confusion matrix for the breathing-based model, highlighting its major shortcoming: an excessive number of false negatives (>200 instances). This implies that the model frequently classified fatigued individuals as alert, which significantly undermines its reliability. In contrast, the heart-based OpenDriver

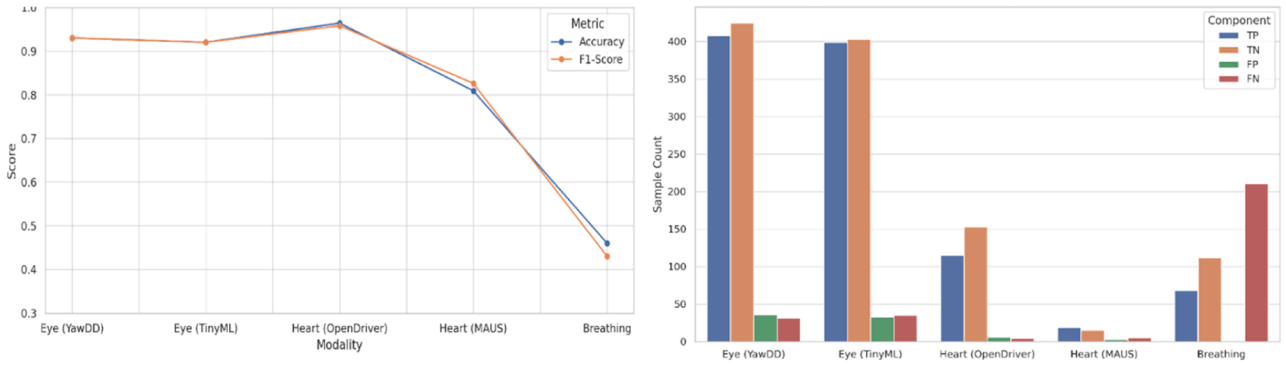


Fig. 29. Line chart: accuracy vs F1 Score across modalities (left), confusion matrix of Random Forest classifier on breathing-based dataset (right).

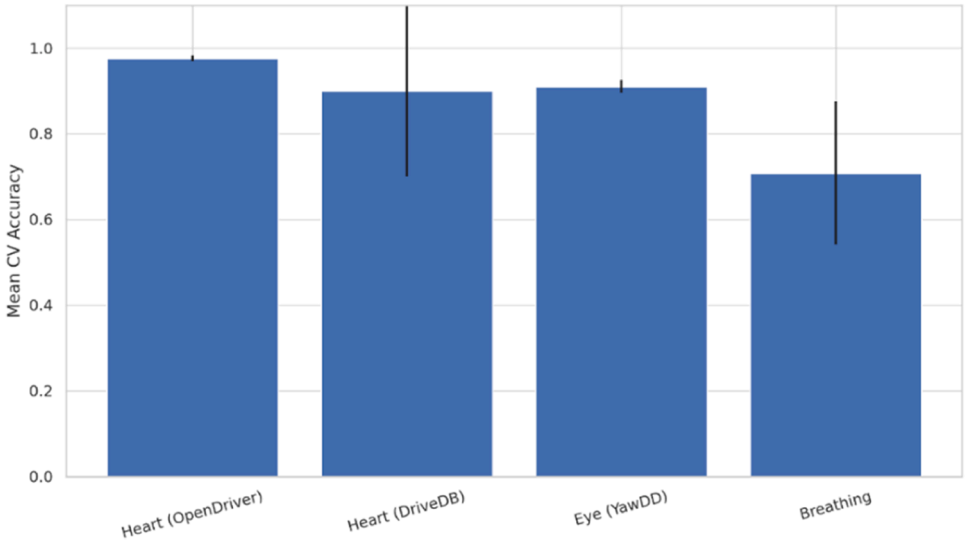


Fig. 30. Bar chart showing mean CV accuracy and standard deviation, highlighting the high stability of heart-based models compared to high variance in breathing-based models.

Table 16
Mean CV accuracy & CV Std Dev.

Modality	Mean CV Accuracy	CV Std Dev
Heart (OpenDriver)	97.6 %	0.73 %
Heart (DriveDB)	90.0 %	20.0 %
Eye (YawDD)	91.0 % (Epoch 20)	-
Breathing	70.8 %	16.7 %

model displayed a highly favorable confusion structure, with minimal false positives and false negatives. This balance affirms its utility in high-stakes fatigue detection environments where both types of error must be minimized.

Cross-validation used to assess model generalizability and training robustness. Fig. 30 illustrates the mean CV accuracy alongside standard deviation across modalities. The heart-based OpenDriver model achieved the most stable results (mean CV accuracy = 97.6 %, SD = 0.73 %), reinforcing its strong generalization. The DriveDB heart model, however, exhibited considerable variability (SD = 20 %), likely due to smaller sample sizes or experimental inconsistencies. Eye-based CNN models also demonstrated acceptable consistency, though standard deviations not reported due to early stopping. The breathing-based model again showed high variance (SD = 16.7 %), suggesting possible overfitting or inter-subject heterogeneity as shown in Table 16.

Table 17
Top influential features by modality.

Modality	Top Feature	Description
Heart	RR Range, Signal Std	Heart rate variability, beat-to-beat change
Breathing	RSA_Gates_SD_Bl	Resp. sinus arrhythmia gate variation
Eye (CNN)	,	Latent features learned from images

6.10.2. Feature importance, interpretability and per-subject fatigue variability

Both the heart- and breathing-based models relied on handcrafted statistical features, allowing for interpretable insights into fatigue physiology. Table 17 summarizes the most influential features by modality. For ECG signals, the most predictive features included RR Range and Signal Standard Deviation, reflecting beat-to-beat variability and signal dispersion. In the breathing model, RSA_Gates_SD_Bl (a measure of respiratory sinus arrhythmia variability) identified as a key indicator of fatigue. In contrast, eye-based CNNs learned latent features through deep representation learning, providing robust performance but limited interpretability, a trade-off often observed in image-based models.

Inter-subject variability in fatigue expression explored using label distribution analysis. Fig. 31 depicts fatigue vs. non-fatigue ratio across subjects in the breathing dataset. Some individuals consistently labeled as fatigued or alert, which, if not properly accounted for during model training, can introduce subject-specific bias. This observation further supports the use of GroupShuffleSplit during data partitioning to

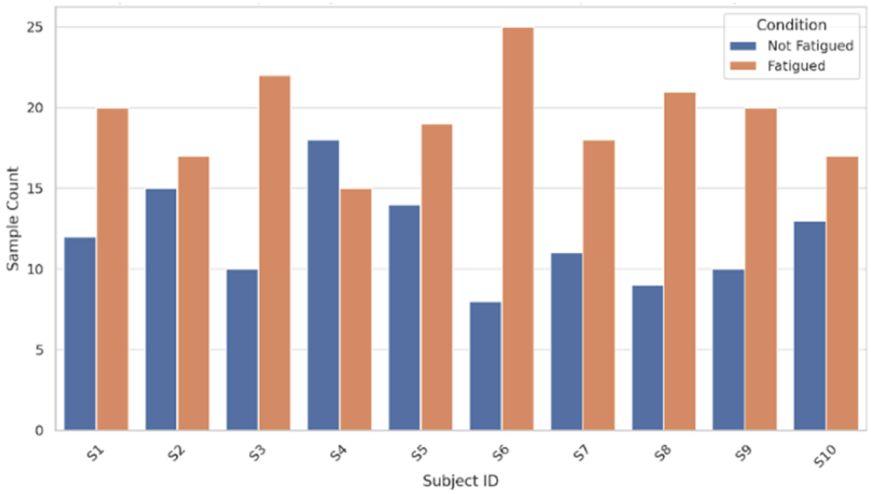


Fig. 31. Per-subject fatigue label distribution (breathing dataset example).

Table 18
Summary of comparative findings.

Aspect	Eye-Based (CNN)	Heart-Based (ECG)	Breathing-Based
Accuracy	High (92–93 %)	Very High (96 %)	Low (46 %)
Sensitivity to Fatigue	Balanced	Excellent	Poor
Interpretability	Low (CNN black box)	High (features interpretable)	High
Real-Time Feasibility	Real-time capable	Needs signal filtering	Fast
Feature Design Required	(CNN auto-learns)	Statistical design	Provided features

prevent data leakage and ensure true generalization.

6.10.3. *Summary of comparative findings*

As summarized in Table 18, the comparative evaluation across key performance and practical dimensions reveals distinct strengths and limitations among the three fatigue detection modalities. The eye-based CNN models consistently achieved high accuracy (92–93 %), demonstrating reliable fatigue detection performance. Their non-intrusive nature and real-time applicability, enabled by simple camera input, make them particularly suited for deployment in intelligent vehicle systems. However, their limited interpretability, due to the latent nature of learned features in convolutional neural networks, remains a trade-off.

The heart-based ECG models outperformed all other approaches in terms of accuracy, reaching up to 96 %, and showed excellent sensitivity to fatigue states. Their use of handcrafted, physiologically grounded statistical features, such as RR variability, allowed for high model interpretability. Nevertheless, they require signal preprocessing and filtering, which may introduce latency or necessitate dedicated hardware in real-time systems. In contrast, the breathing-based classifier, despite leveraging a rich set of precomputed features, exhibited substantially lower performance, with an accuracy of only 46 % and poor recall. While its features offered high interpretability and fast computation, the model's reduced sensitivity to fatigue renders, it unsuitable for standalone use in safety-critical contexts. These findings highlight the superior reliability and robustness of eye and heart modalities, both in terms of classification performance and real-world feasibility. The breathing modality, while interpretable, appears to lack sufficient discriminatory power, further reinforcing the value of multi-modal fusion strategies in future fatigue detection systems.

Comparative evaluation underscores the superior performance of eye- and heart-based modalities in fatigue detection tasks. Both approaches achieved high precision, recall, and accuracy, exceeding 92 % in most metrics, making them well-suited for deployment in safety-critical environments. Heart-based models, with interpretable features and physiological grounding, offer strong diagnostic potential but require careful signal acquisition and preprocessing. Eye-based CNNs, on the other hand, provide a practical solution with minimal invasiveness and strong generalizability using only camera input. Conversely, the breathing-based model underperformed across multiple evaluation dimensions. Its low recall rate, coupled with high variability in cross-validation, suggests a weaker association between respiratory patterns and fatigue, at least within the context of the available features. These limitations render it unsuitable as a standalone fatigue detection solution but leave room for improvement through temporal modeling or multi-modal fusion. The collective findings point toward multi-modal integration, particularly combinations of eye and heart data, as a promising direction for robust, real-time, and explainable fatigue monitoring systems in vehicle or workplace settings.

6.11. *Real-Time driver health monitoring via IoT and cloud integration*

This section presents the outcomes of the cloud-integrated system designed for real-time driver health monitoring. The system was evaluated based on data upload latency, anomaly detection accuracy, and responsiveness of alert mechanisms under varying driving conditions. The results demonstrate the robustness, scalability, and practicality of integrating physiological sensing with Azure Cloud and IoT infrastructure. The system was tested in four distinct driving conditions, calm driving, heavy traffic, night driving, and emergency braking, to evaluate its responsiveness and accuracy in real-world use cases. The results summarized in Table 1. The system maintained a low data upload latency across all test cases, with the lowest (150 ms) recorded during emergency braking scenarios, where system reactivity is most critical. Alert responsiveness ranged from 220 to 300 ms, demonstrating near real-time communication between edge devices, the cloud, and alerting endpoints. Detection accuracy remained above 90 % in all cases, peaking at 95 % during calm driving. While there was a slight dip in accuracy (92 %) under heavy traffic conditions due to increased data volume and minor transmission delays, overall detection reliability remained high. The integration with Azure Power BI dashboards enabled live monitoring of driver biosignals (heart rate, eye movement, breathing rate) as shown in Fig. 23. These dashboards provided both raw data streams and computed anomaly states in real time. Detected anomalies, including potential fatigue events or critical health deviations, were clearly visualized with temporal markers and alert classifications.

The pre-trained Isolation Forest model, deployed via Azure Machine

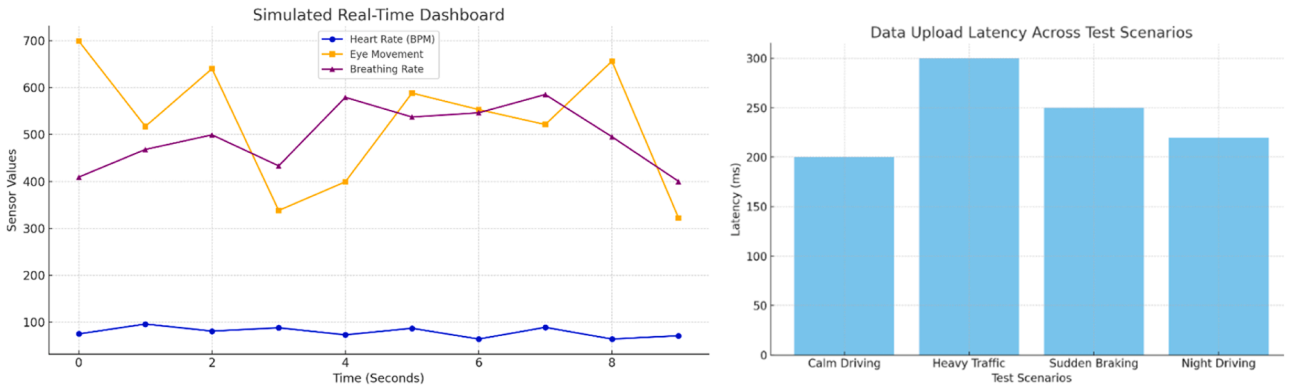


Fig. 32. Real-time data visualization dashboard (left), Latency metrics across driving scenarios (right).

Learning Services, processed incoming feature vectors from the Arduino devices with minimal latency. Fatigue-related anomalies flagged with high precision, enabling immediate intervention. Alert Delivery System: Alerts triggered by cloud inference pushed to local vehicle interfaces (e.g., buzzers, LEDs) and remote dashboards accessible by fleet managers. As illustrated in Fig. 32 (left), the data flow from sensor input to alert output demonstrated high throughput and reliability. As visualized in Fig. 32 (right), latency plotted across various test scenarios. The system exhibited graceful degradation under high-load conditions, such as heavy traffic but maintained acceptable thresholds (<300 ms), ensuring that safety-critical interventions remained timely. These results validate the effectiveness of combining low-cost IoT sensors, cloud-based analytics, and real-time feedback mechanisms to support driver safety and health monitoring. The system's high detection accuracy, fast data processing, and robust visualization tools position it as a viable solution for scalable deployment in smart mobility platforms.

7. Conclusion

This study presented a scalable driver health monitoring framework integrating multimodal physiological sensing, machine learning based fatigue detection, and real-time cloud infrastructure. The system was designed to support proactive and continuous monitoring of driver well-being within connected and electric vehicle environments. By combining ECG signals, eye image analysis, and respiratory measurements, the framework captures complementary physiological indicators associated with fatigue and enables real-time physiological data integration and fatigue-related pattern analysis. Experimental results demonstrate that the ECG-based Random Forest model achieves strong classification performance, particularly on real-world datasets such as OpenDriver, with consistent behavior across DriveDB and MAUS. The CNN-based eye fatigue model also delivers high accuracy under varying lighting and image quality conditions, confirming the reliability of visual fatigue cues. Although the respiration-based component exhibits limited standalone sensitivity due to sensor resolution and feature overlap, it provides contextual information during high-stress driving scenarios and may serve as a complementary modality in multimodal configurations. A key contribution of this work is the end-to-end cloud IoT integration, where physiological data collected at the edge are securely transmitted and processed in near real time using Azure IoT Hub and Stream Analytics. The system maintains low-latency performance (below 300 ms) across evaluated scenarios, and real-time dashboards enable visualization of driver health states, fatigue trends, and anomaly events. While the current framework demonstrates promising performance, limitations such as heuristic fatigue labeling, lack of personalization, and reliance on simulated sensors indicate directions for future work, including adaptive modeling, clinical-grade sensor validation, and larger-scale evaluations.

CRediT authorship contribution statement

Obaida AlHousrya: Conceptualization, Methodology, Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Aseel Bennagi:** Methodology, Validation, Writing – original draft, Writing – review & editing. **Petru A. Cotfas:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Daniel T. Cotfas:** Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This study uses multiple publicly available datasets (OpenDriver, DriveDB, MAUS, YawDD, TinyML, and the Driver Respiration Dataset). All dataset URLs and references are provided in the manuscript.

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